Chemistry

Class 12 (English Medium)

Solved Exercises — Sindh Textbook Board

13 chapters — 367 questions

Chapter 1: Chemistry of Representative Elements

Multiple Choice Questions

1. (i) Melting and boiling points regularly decrease down the group in:

✓ (a) Group IA

(b) Group IIIA

(c) Group VIIA

(d) Group VIIIA

Answer: (a) Group IA — In alkali metals the metallic bonding weakens regularly as atomic size increases down the group, so melting and boiling points decrease steadily from Li to Cs. In the other groups the trend is irregular because of changes in structure and bonding type.

2. (ii) The s-block element forms super oxide when burned in air is:

(a) Li

(b) Na

✓ (c) K

(d) Mg

Answer: (c) K — Potassium is large and reactive enough to form the superoxide KO_2 on burning in excess air. Lithium forms only the normal oxide Li_2O and sodium mainly forms the peroxide Na_2O_2.

3. (iii) The general formula of nitrides for alkaline earth metal is:

(a) MN_3

(b) M_2N_3

(c) M_3N

✓ (d) M_3N_2

Answer: (d) M3N2 — An alkaline earth metal forms M2+ ions and nitrogen forms N3- ions, so three M2+ ions balance two N3- ions. Examples are Mg3N2 and Ca3N2, formed when the metal is heated in nitrogen.

4. (iv) The alkali metal that gives a yellow color during a flame test is:

✓ (a) Na

(b) K

(c) Rb

(d) Cs

Answer: (a) Na — Sodium salts colour the flame golden yellow because the excited sodium atoms emit light of this colour on returning to the ground state. Potassium gives a lilac flame, while rubidium and caesium give red-violet and blue-violet flames respectively.

5. (v) The chemical commonly used in fireworks is:

(a) Sodium bicarbonate

(b) Bleaching powder

✓ (c) Potassium nitrate

(d) Potash Alum

Answer: (c) Potassium nitrate — Potassium nitrate (saltpetre) is the oxidising agent in gunpowder-type firework mixtures and supplies oxygen for rapid burning. Metal salts present with it produce the characteristic colours of fireworks.

6. (vi) Cathode in Castner Kellner cell is:

(a) Titanium blocks

(b) Carbon rods

✓ (c) Mercury

(d) Iron container

Answer: (c) Mercury — In the Castner-Kellner cell a flowing layer of mercury acts as the cathode, where sodium ions are discharged to form sodium amalgam. The amalgam is then decomposed with water in a separate compartment to give pure NaOH and hydrogen.

7. (vii) The diagonal member of beryllium is:

(a) Mg

✓ (b) Al

(c) Si

(d) C

Answer: (b) Al — Beryllium lies diagonally next to aluminium in the periodic table and resembles it closely in properties such as amphoteric oxides and hydroxides. This similarity is an example of diagonal relationship.

Question: 8. (viii) The catalyst commonly used in the conversion of sulfur dioxide to sulfur trioxide during the preparation of sulphuric acid is: (a) Vanadium pentoxide (V2O5) (b) Copper (Cu) (c) Iron (Fe) (d) Nickel (Ni)

Answer: (a) Vanadium pentoxide (V2O5) — In the Contact process, 2SO2 + O2 is converted to 2SO3 at about 450 C in the presence of V2O5 catalyst. This is the key step in the manufacture of sulphuric acid.

9. (ix) Oil of vitriol refers to:

(a) Borax

✓ (b) Sulphuric acid

(c) Alum

(d) Caustic soda

Answer: (b) Sulphuric acid — Concentrated sulphuric acid is an oily, dense liquid and was historically called oil of vitriol. It is obtained from vitriol minerals, which is the origin of this old name.

10. (x) The best oxidizing agent among halogens is:

(a) F2

(b) Cl2

(c) Br2

(d) I2

Answer: (a) F2 — Fluorine has the highest electronegativity and the greatest tendency to gain electrons, so it is the strongest oxidising agent among the halogens. Oxidising power decreases down the group in the order F2 > Cl2 > Br2 > I2.

Short Questions

Question: 1(i). Give reasons for the following: Ionization energy decreases from top to bottom in s-block elements?

Answer: Down the group a new shell is added at each step, so the valence electron is farther from the nucleus and is better shielded by the inner electrons. The effective nuclear attraction on it therefore decreases, and less energy is needed to remove it, so ionization energy falls from top to bottom.

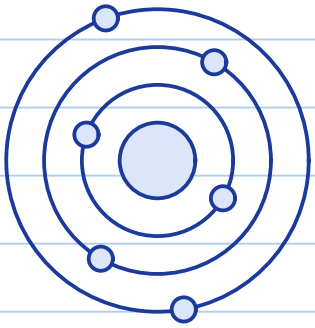


Fig.: Atom (labelled diagram)

Question: 1(ii). Give reasons for the following: Boiling point of halogens increase down the group in the periodic table?

Answer: Down the group the halogen molecules become larger and their electron clouds become more easily polarised. The van der Waals forces between the molecules therefore become stronger, more heat is needed to separate them, and boiling point increases from F2 to I2.

Question: 1(iii). Give reasons for the following: Gallium has smaller atomic radii than aluminum despite being below the aluminum in group IIIA?

Answer: Gallium comes after the first transition series, and its ten 3d electrons shield the nuclear charge poorly. The increased effective nuclear charge pulls the outer electrons closer, a contraction called the d-block contraction, so gallium is slightly smaller than aluminium.

Question: 1(iv). Give reasons for the following: Electronegativities of alkali metals decrease from Li to Cs?

Answer: Down the group the atomic size increases and the valence electron is held less tightly because of greater distance and shielding. The atom therefore attracts a shared pair of electrons less strongly, so electronegativity decreases from Li to Cs.

Question: 1(v). Give reasons for the following: Alkali metals are good conductor of electricity?

Answer: Alkali metals have one loosely held valence electron per atom in a sea of delocalised electrons within the metallic lattice. These free electrons move readily under an applied potential, so the metals conduct electricity and heat very well.

Question: 1(vi). Give reasons for the following: Acidity of hydrogen halides increase from HF to HI?

Answer: Down the group the H-X bond becomes longer and weaker as the halogen atom becomes larger, so the bond breaks more easily to release H+ ions. Bond strength, and not electronegativity, controls the acidity here, so the order is HI > HBr > HCl > HF.

Question: 1(vii). Give reasons for the following: Fluorine is the strongest oxidizing agent?

Answer: Fluorine is the smallest and most electronegative halogen, and the F-F bond is weak while the hydration energy of the small F- ion is very high. It therefore gains electrons most readily and oxidises other substances most powerfully, making F2 the strongest oxidising agent.

Question: 2. What is flame test? Mention the colour of flame of alkali metals.

Answer: In a flame test a small amount of a metal salt on a clean platinum or nichrome wire is held in a Bunsen flame, and the metal imparts a characteristic colour to the flame. The colours are: lithium crimson red, sodium golden yellow, potassium lilac, rubidium red-violet and caesium blue-violet. The test is used to identify alkali metal ions in a sample.

Question: 3. Explain the auto oxidizing and reducing properties of chlorine.

Answer: In some reactions chlorine is oxidised and reduced at the same time, a process called disproportionation, so it acts as its own oxidising and reducing agent. With cold dilute NaOH it gives chloride and hypochlorite: Cl2 + 2NaOH -> NaCl + NaOCl + H2O, in which chlorine goes from 0 to -1 and to +1. With hot concentrated NaOH it forms chloride and chlorate: 3Cl2 + 6NaOH -> 5NaCl + NaClO3 + 3H2O.

Question: 4. What is meant by a diagonal relationship? Mention three pairs of representative elements that show diagonal relationship.

Answer: A diagonal relationship is the similarity in properties between an element and the element lying diagonally below it in the next group, caused by a balance of increasing size down the group and increasing charge across the period. Three such pairs are lithium and magnesium, beryllium and aluminium, and boron and silicon. For example, beryllium and aluminium both form amphoteric oxides and covalent chlorides.

Question: 5. Discuss the group trend of Ionization energy in group IIIA of periodic table.

Answer: Ionization energy in group IIIA does not decrease regularly: it falls from B to Al, then rises slightly at Ga, falls at In and rises again at Tl. The rises at Ga and Tl are due to the poor shielding of the filled d and f electrons (d-block and f-block contractions), which increases the effective nuclear charge. Thus the overall trend is a decrease with irregular increases at gallium and thallium.

Question: 6. Write down three properties of beryllium that show its unique behavior in group IIA.

Answer: Beryllium differs from the other group IIA metals in three clear

ways. First, its oxide and hydroxide are amphoteric and dissolve in both acids and alkalis, unlike the basic oxides of the others. Second, its compounds are largely covalent and it never shows a coordination number greater than four. Third, it does not react with water even on heating, whereas magnesium and the heavier metals react readily.

Descriptive Questions

Question: 1. Draw a flow diagram of contact process and describe various steps involved in the manufacturing of sulphuric acid.

Answer: The flow is: sulphur (or iron pyrites) burner -> SO2 gas -> purification unit -> converter containing V2O5 at about 450 C -> SO3 gas -> absorption tower with concentrated H2SO4 -> oleum -> dilution with water -> concentrated H2SO4. Step 1, burning: S + O2 -> SO2. Step 2, catalytic oxidation: 2SO2 + O2 -> 2SO3 in the presence of V2O5. Step 3, absorption: SO3 + H2SO4 -> H2S2O7 (oleum), followed by H2S2O7 + H2O -> 2H2SO4, because direct absorption of SO3 in water produces a dense mist.

Question: 2. Explain with the help of a diagram of Castner Kellner cell, how caustic soda is obtained by the electrolysis of aqueous sodium chloride?

Answer: The cell has a flowing mercury cathode at the bottom and titanium anode blocks in brine compartments separated by partitions, with a central compartment containing water or dilute NaOH. At the anode, chloride ions are discharged: 2Cl- -> Cl2 + 2e-, and chlorine gas is collected. At the mercury cathode, sodium ions are discharged into the mercury: Na+ + e- -> Na (amalgam). The sodium amalgam flows to the central compartment where it reacts with water: 2Na(Hg) + 2H2O -> 2NaOH + H2 + Hg, and the mercury is recycled, so pure caustic soda free

from NaCl is obtained.

Question: 3(i). Write the balance equations for the following chemical process. (i) A piece of aluminum is dipped into concentrated sulphuric acid.

Answer: Aluminium is oxidised by hot concentrated sulphuric acid, which itself is reduced to sulphur dioxide. The balanced equation is: 2Al + 6H2SO4 -> Al2(SO4)3 + 3SO2 + 6H2O. This shows that concentrated H2SO4 acts as a strong oxidising agent towards metals.

Question: 3(ii). Write the balance equations for the following chemical process. (ii) Ferric chloride is mixed with an aqueous solution of caustic soda.

Answer: A brown precipitate of ferric hydroxide is formed in this double decomposition reaction. The balanced equation is: FeCl3 + 3NaOH -> Fe(OH)3 + 3NaCl. The reddish-brown Fe(OH)3 precipitate is a characteristic test for Fe3+ ions.

Question: 3(iii). Write the balance equations for the following chemical process. (iii) Sodium burns in excess of air.

Answer: In excess air sodium burns to form mainly sodium peroxide, a pale yellow solid. The balanced equation is: 2Na + O2 -> Na2O2. With a limited supply of air the normal oxide is formed instead: 4Na + O2 -> 2Na2O.

Question: 3(iv). Write the balance equations for the following chemical process. (iv) Magnesium is heated with nitrogen gas.

Answer: Hot magnesium combines directly with nitrogen to form magnesium nitride, a greenish-yellow solid. The balanced equation is: 3Mg

+ N2 -> Mg3N2. The nitride is hydrolysed by warm water to give ammonia: Mg3N2 + 6H2O -> 3Mg(OH)2 + 2NH3.

Question: 3(v). Write the balance equations for the following chemical process. (v) Potassium is put into ethyl alcohol.

Answer: Potassium reacts with ethanol to form potassium ethoxide with the evolution of hydrogen gas, more vigorously than sodium does. The balanced equation is: 2K + 2C2H5OH -> 2C2H5OK + H2. This reaction shows the weakly acidic character of the -OH group of alcohols.

Question: 3(vi). Write the balance equations for the following chemical process. (vi) Chlorine gas is passed through an aqueous solution of caustic soda.

Answer: With cold dilute caustic soda, chlorine disproportionates to sodium chloride and sodium hypochlorite: Cl2 + 2NaOH -> NaCl + NaOCl + H2O. With hot concentrated caustic soda the products are sodium chloride and sodium chlorate: 3Cl2 + 6NaOH -> 5NaCl + NaClO3 + 3H2O. In both cases chlorine acts as both oxidising and reducing agent.

Question: 4. What is bleaching powder? How it is prepared? Give the reaction of bleaching powder with water and hydrochloric acid.

Answer: Bleaching powder is calcium oxychloride, Ca(OCl)Cl, a yellowish-white powder used as a bleaching and disinfecting agent. It is prepared by passing chlorine gas over dry slaked lime at about 40 C: Ca(OH)2 + Cl2 -> Ca(OCl)Cl + H2O. With water it gives nascent oxygen which does the bleaching: Ca(OCl)Cl + H2O -> Ca(OH)2 + Cl2 in presence of moisture and CO2 it slowly releases chlorine, and with dilute hydrochloric acid it liberates chlorine rapidly: Ca(OCl)Cl + 2HCl -> CaCl2 +

$Cl_2 + H_2O$.

Question: 5. Discuss the group trend of atomic radii, ionization energy and electronegativity of alkali metals.

Answer: Atomic radii increase regularly from Li to Cs because a new electron shell is added at each step and shielding increases. Ionization energy decreases from Li to Cs, as the valence electron is farther from the nucleus and more shielded, so it is removed more easily. Electronegativity also decreases from Li (1.0) to Cs (0.7) for the same reason: the larger atom attracts a shared pair of electrons less strongly. As a result, metallic and electropositive character, and chemical reactivity, increase down the group.

Chapter 2: Chemistry of Outer Transition (d-Block)

Elements

Multiple Choice Questions

1. (i) Zn+2 ion is colourless because:

(a) Its undergoes d-d transition of electron

(b) Its 3d orbitals have all unpaired electrons

✓ (c) Its 3d orbitals have all paired electrons

(d) Its d orbitals split up into t2g and eg

Answer: (c) Its 3d orbitals have all paired electrons — Zn2+ has the configuration 3d10, so all its d electrons are paired and no d-d transition is possible. It therefore absorbs no visible light and its compounds are white or colourless.

2. (ii) The coordination number of cobalt in Na4[Co(C2O4)3] is:

(a) 3

(b) 4

✓ (c) 6

(d) 7

Answer: (c) 6 — Oxalate, C2O4 2-, is a bidentate ligand and donates two electron pairs per ligand. Three oxalate ligands therefore form six coordinate bonds with cobalt, giving a coordination number of 6.

3. (iii) An example of a bidentate ligand among the following is:

(a) OH-

✓ (b) C2O4-2

(c) Cl-

(d) CN-

Answer: (b) C2O4-2 — The oxalate ion has two oxygen donor atoms through which it binds to the metal at two positions, forming a ring. OH-, Cl- and CN- each donate through only one atom and are monodentate ligands.

4. (iv) A highly paramagnetic ion among the following is:

(a) Fe+2

✓ (b) Fe+3

(c) Co+2

(d) Cr+3

Answer: (b) Fe+3 — Fe3+ has the configuration 3d5 with five unpaired electrons, the maximum among the given ions. Paramagnetism increases with the number of unpaired electrons, so Fe3+ is the most strongly paramagnetic of these ions.

5. (v) The highest oxidation state of chromium is:

(a) +4

(b) +5

✓ (c) +6

(d) +7

Answer: (c) +6 — Chromium (3d5 4s1) can use all six valence electrons in bonding, so its highest oxidation state is +6. It is shown in compounds such as K2Cr2O7 and CrO3, where chromium is a powerful oxidising agent.

6. (vi) The element is not used for electroplating:

(a) Zinc

(b) Tin

(c) Chromium

✓ (d) Manganese

Answer: (d) Manganese — Zinc, tin and chromium are all commonly deposited by electroplating to protect or decorate metal surfaces. Manganese is too reactive and difficult to deposit from aqueous solution, so it is not used for electroplating.

7. (vii) The steel is typically used in making Fry pans:

(a) Carbon steel

✓ (b) Stainless steel

(c) Tool Steel

(d) Alloy Steel

Answer: (b) Stainless steel — Stainless steel contains iron with chromium and nickel, so it resists rusting, does not react with food and is easy to clean. These properties make it the usual material for fry pans and other kitchen utensils.

8. (viii) The step which involved in the extraction of copper from chalcopyrite ore in the elimination of gangue impurities is:

✓ (a) Concentration

(b) Roasting

(c) Smelting

(d) Bessemerization

Answer: (a) Concentration — The ore is first concentrated, usually by froth flotation, in which the gangue (earthy impurities) is removed and the copper mineral is enriched. Roasting, smelting and Bessemerization are later steps that act on the already concentrated ore.

9. (ix) 5d series of outer transition elements is:

(a) Sc to Zn

(b) Y to Cd

(c) La to Hg

(d) Ac to Cn

Answer: (c) La to Hg — The 5d series lies in the sixth period and runs from lanthanum (La) to mercury (Hg). Sc to Zn is the 3d series, Y to Cd is the 4d series and Ac to Cn is the 6d series.

10. (x) Oxidation of manganese in air gives the following oxide:

(a) MnO

(b) MnO3

(c) Mn2O3

(d) Mn3O4

Answer: (d) Mn3O4 — When manganese is heated or oxidised in air it forms the mixed oxide Mn3O4 (a mixture of MnO and Mn2O3). The reaction is: 3Mn + 2O2 -> Mn3O4.

Short Questions

Question: 1(i). Write the IUPAC names of the following: (i) Na2[Pt(OH)4]

Answer: The complex anion contains Pt with four hydroxo ligands; the charge balance with two Na+ ions shows Pt is in the +2 oxidation state. Its IUPAC name is sodium tetrahydroxoplatinate(II). The name of the anionic complex ends in -ate and the oxidation state is shown in Roman numerals.

Question: 1(ii). Write the IUPAC names of the following: (ii) K2[Fe(CN)5NO]

Answer: This is the nitroprusside ion, containing five cyano ligands and one nitrosyl (NO) ligand around iron(II). Its IUPAC name is potassium

pentacyanonitrosylferrate(II). The ligands are named alphabetically and the anionic complex of iron takes the name ferrate.

Question: 1(iii). Write the IUPAC names of the following: (iii) [Zn(NH3)4]+2

Answer: The cation contains four ammine (NH3) ligands around zinc(II), and the complex is cationic so the metal keeps its ordinary name. Its IUPAC name is tetraamminezinc(II) ion. Zinc shows only the +2 oxidation state in its complexes.

Question: 1(iv). Write the IUPAC names of the following: (iv) [Ni(SCN)4]-2

Answer: The anion contains four thiocyanato ligands around nickel(II), and an anionic complex of nickel is named as a nickelate. Its IUPAC name is tetrathiocyanatonickelate(II) ion. The -ate ending and the Roman numeral II follow the rules for anionic complexes.

Question: 2(i). Give reasons for the following: (i) Why do transition elements show variable oxidation states?

Answer: In transition elements the (n-1)d and ns orbitals have very similar energies, so a variable number of electrons from both can take part in bonding. Different numbers of electrons are lost or shared under different conditions, giving several oxidation states. For example, manganese shows states from +2 to +7.

Question: 2(ii). Give reasons for the following: (ii) Why transition elements have ability to form alloys?

Answer: Transition elements have very similar atomic sizes, differing only slightly across a series, so atoms of one metal can easily replace atoms of another in the lattice without distorting it much. They therefore mix readily

in the molten state and form solid solutions called alloys, such as brass, bronze and stainless steel.

Question: 2(iii). Give reasons for the following: (iii) Why Cu+2 ion is blue but Zn+2 is colourless?

Answer: Cu2+ has the configuration 3d9 with one unpaired electron, which can absorb light of a particular colour and be promoted within the split d orbitals (d-d transition); the transmitted light appears blue. Zn2+ is 3d10 with all electrons paired, so no d-d transition is possible and its compounds are colourless.

Question: 2(iv). Give reasons for the following: (iv) Why chromium exists in 4s1 3d5 configuration but not in 4s2 3d4?

Answer: A half-filled d subshell (3d5) has extra stability because of symmetrical electron distribution and maximum exchange energy. One electron therefore shifts from 4s to 3d to make the d subshell exactly half-filled. This makes 4s1 3d5 lower in energy and more stable than 4s2 3d4 for chromium.

Question: 2(v). Give reasons for the following: (v) Why binding energy of zinc is least in 3d series?

Answer: Binding (cohesive) energy of a metal depends on the number of unpaired d electrons available for metallic bonding. Zinc has a completely filled 3d10 configuration with no unpaired d electrons, so it contributes least to metallic bonding. Consequently its binding energy, melting point and hardness are the lowest in the 3d series.

Question: 3(i). Write down the balanced chemical equations for the

following reactions. (i) Reaction of conc. nitric acid with copper

Answer: Concentrated nitric acid oxidises copper, itself being reduced to nitrogen dioxide. The balanced equation is: Cu + 4HNO3 -> Cu(NO3)2 + 2NO2 + 2H2O. Brown fumes of NO2 are evolved and a blue solution of copper nitrate is formed.

Question: 3(ii). Write down the balanced chemical equations for the following reactions. (ii) Reaction of conc. sulphuric acid with copper

Answer: Hot concentrated sulphuric acid oxidises copper and is reduced to sulphur dioxide. The balanced equation is: Cu + 2H2SO4 -> CuSO4 + SO2 + 2H2O. Copper does not react with dilute sulphuric acid because it lies below hydrogen in the reactivity series.

Question: 3(iii). Write down the balanced chemical equations for the following reactions. (iii) Reaction of permanganate with oxalic acid

Answer: In acid solution permanganate oxidises oxalic acid to carbon dioxide and is itself reduced to colourless Mn2+. The balanced equation is: 2KMnO4 + 5H2C2O4 + 3H2SO4 -> K2SO4 + 2MnSO4 + 10CO2 + 8H2O. The purple colour of KMnO4 is discharged, which is used in volumetric analysis.

Question: 3(iv). Write down the balanced chemical equations for the following reactions. (iv) Reaction of dichromate with ferrous sulphate

Answer: In acid solution dichromate oxidises Fe2+ to Fe3+ and is itself reduced to green Cr3+. The balanced equation is: K2Cr2O7 + 6FeSO4 + 7H2SO4 -> K2SO4 + Cr2(SO4)3 + 3Fe2(SO4)3 + 7H2O. The orange colour of dichromate changes to green during the reaction.

Question: 3(v). Write down the balanced chemical equations for the following reactions. (v) Reaction of manganese with dilute sulphuric acid

Answer: Manganese lies above hydrogen in the reactivity series, so it dissolves in dilute sulphuric acid with evolution of hydrogen. The balanced equation is: Mn + H2SO4 -> MnSO4 + H2. A pale pink solution of manganese sulphate is obtained.

Question: 3(vi). Write down the balanced chemical equations for the following reactions. (vi) Reaction of iron with chlorine

Answer: Iron burns in chlorine to form ferric chloride, because chlorine is a strong oxidising agent and takes iron to its higher oxidation state. The balanced equation is: 2Fe + 3Cl2 -> 2FeCl3. Brown fumes of FeCl3 are produced.

Question: 4. Why d-block elements are called outer transition elements?

Answer: They are called transition elements because their properties are intermediate between those of the highly reactive s-block metals and the p-block elements, and they form a bridge between them in the periodic table. They are called outer transition elements because their differentiating electron enters the outer (n-1)d subshell, in contrast to the inner transition (f-block) elements. Their atoms or ions have partially filled d orbitals, which gives them their characteristic properties.

Question: 5. Write down the effect of pH changes on dichromate equilibrium in water.

Answer: In aqueous solution chromate and dichromate exist in equilibrium: 2CrO4 2- (yellow) + 2H+ <-> Cr2O7 2- (orange) + H2O. Adding acid (lowering pH) shifts the equilibrium to the right and the solution turns

orange due to dichromate. Adding alkali (raising pH) removes H+ ions, shifts the equilibrium to the left, and the solution becomes yellow due to chromate.

Question: 6. Melting point of d-block elements increase up to middle of the series and then decrease why?

Answer: Melting point depends on the strength of metallic bonding, which increases with the number of unpaired d electrons taking part in bonding. Up to the middle of the series (Cr, Mo, W group) the number of unpaired electrons is maximum, so bonding is strongest and melting point is highest. Beyond the middle, electrons begin to pair up, metallic bonding weakens, and melting point decreases towards zinc, cadmium and mercury.

Question: 7. Give the composition and applications of stainless steel, brass and bronze.

Answer: Stainless steel contains about 74% iron, 18% chromium and 8% nickel; it is used for cutlery, surgical instruments, fry pans and chemical plant because it resists rusting. Brass contains about 70% copper and 30% zinc; it is used for utensils, taps, locks and musical instruments. Bronze contains about 90% copper and 10% tin; it is used for statues, coins, bells and bearings because it is hard and resists corrosion.

Descriptive Questions

Question: 1(a). Explain the trend of following properties of 3d-series of transition elements. (a) Paramagnetic behavior

Answer: Paramagnetism is caused by unpaired electrons, and its strength increases with their number. Across the 3d series the number of unpaired

electrons rises from Sc (1) to a maximum at Mn/Fe region (Mn2+ and Fe3+ have 5) and then falls as pairing begins, reaching zero at Zn (3d10). Hence paramagnetism first increases, becomes maximum in the middle of the series, and then decreases; Zn2+ and Sc3+ are diamagnetic.

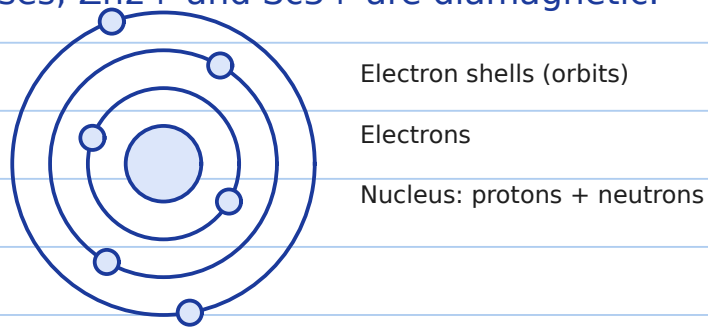


Fig.: Atom (labelled diagram)

Question: 1(b). Explain the trend of following properties of 3d-series of transition elements. (b) Variable oxidation state

Answer: Transition elements show variable oxidation states because the (n-1)d and ns electrons are close in energy and both can be used in bonding. In the 3d series the common state is +2 (loss of 4s electrons), and the maximum oxidation state rises from Sc (+3) to Mn (+7), equal to the total number of 4s and 3d electrons. After manganese the maximum falls (Fe +6, Co +5, Ni +4, Cu +2, Zn +2) because the d electrons become held more tightly.

Question: 1(c). Explain the trend of following properties of 3d-series of transition elements. (c) Colour formation

Answer: Most transition metal compounds are coloured because their ions have partially filled d orbitals which split into two sets (t2g and eg) in the presence of ligands. Electrons absorb visible light of a definite colour to jump between these sets (d-d transition), and the compound shows the complementary colour. Ions with d0 (Sc3+, Ti4+) or d10 (Zn2+, Cu+) configurations cannot show d-d transitions and are colourless.

Question: 2. How can you define a coordination complex and a chelating

ligand? Explain various types of ligands with examples.

Answer: A coordination complex is a compound in which a central metal atom or ion is bonded to a fixed number of surrounding ions or molecules (ligands) by coordinate bonds, for example [Cu(NH3)4]2+. A chelating ligand is a polydentate ligand that binds through two or more donor atoms at the same time and forms a ring structure with the metal; the resulting complex is called a chelate, for example oxalate and EDTA complexes. Ligands are classified by the number of donor atoms: monodentate ligands donate through one atom (NH3, Cl-, CN-), bidentate through two atoms (oxalate C2O4 2-, ethylenediamine), and polydentate through many atoms (EDTA is hexadentate). They may also be neutral (NH3, H2O, CO) or negatively charged (Cl-, CN-, C2O4 2-).

Question: 3. Describe how 99.99% pure copper is obtained from its chalcopyrite ore.

Answer: Chalcopyrite, CuFeS2, is first concentrated by froth flotation to remove gangue. The concentrate is roasted in a reverberatory furnace: 2CuFeS2 + O2 -> Cu2S + 2FeS + SO2, and the roasted ore is smelted with silica so that FeO is removed as slag (FeO + SiO2 -> FeSiO3), leaving copper matte (Cu2S + FeS). In the Bessemer converter, air is blown through the matte: 2Cu2S + 3O2 -> 2Cu2O + 2SO2 and Cu2S + 2Cu2O -> 6Cu + SO2, giving blister copper of about 98% purity. Finally it is refined electrolytically using thick plates of blister copper as anode and thin sheets of pure copper as cathode in acidified CuSO4 solution; pure copper (99.99%) deposits on the cathode while impurities collect as anode mud.

Question: 4. Explain why transition elements and their compounds serve as catalysts in many chemical reactions.

Answer: Transition elements act as catalysts because they can show several oxidation states, accept and donate electrons easily, and provide a suitable surface for the reactants to be adsorbed. Reactants form temporary bonds with the metal surface or with the metal ion, which lowers the activation energy of the reaction. Examples are Fe in the Haber process for ammonia, V_2O_5 in the Contact process, Ni in hydrogenation of oils and Pt in catalytic converters.

Question: 5. What is meant by binding energy? Write down the trend of binding energy in 3d series of transition elements.

Answer: Binding energy (cohesive energy) is the energy required to break one mole of a metal into isolated gaseous atoms; it measures the strength of metallic bonding. In the 3d series it increases from Sc to a maximum around V-Cr-Mn in the middle, because the number of unpaired d electrons available for bonding increases. It then decreases as electron pairing begins, and is least for zinc, whose filled $3d^{10}$ shell contributes no unpaired electrons to metallic bonding.

Chapter 3: Organic Compounds

Multiple Choice Questions

1. (i) The functional group that contains a nitrogen atom is:

(a) Alcohol

(b) Ketone

(c) Ester

✓ (d) Amide

Answer: (d) Amide — The amide group -CONH2 contains a nitrogen atom bonded to a carbonyl carbon. Alcohol (-OH), ketone (C=O) and ester (-COO-) groups contain only carbon, hydrogen and oxygen.

2. (ii) The compound belongs to the alkene family is:

(a) CH4

(b) C2H6

(c) C3H8

✓ (d) C2H4

Answer: (d) C2H4 — Alkenes have the general formula CnH2n, and C2H4 (ethene) fits it exactly with a carbon-carbon double bond. CH4, C2H6 and C3H8 fit the alkane formula CnH2n+2 and are saturated compounds.

3. (iii) Destructive distillation of coal gives coke, coaltar, coal gas and:

(a) Natural gas

(b) Petroleum

(c) Naphtha

✓ (d) Ammonia liquor

Answer: (d) Ammonia liquor — When coal is heated strongly in the

absence of air it breaks down into coke, coal tar, coal gas and ammoniacal liquor. The ammonia liquor is the aqueous solution containing ammonia formed from the nitrogen present in coal.

4. (iv) The knocking of internal combustion engine can be reduced by the following petroleum process:

✓ (a) Reforming

(b) Refining

(c) Distillation

(d) Condensation

Answer: (a) Reforming — Reforming converts straight-chain alkanes of petrol into branched-chain and aromatic hydrocarbons, which have a higher octane number and burn smoothly. This reduces knocking, which is the damaging metallic sound produced by uneven combustion in the engine.

5. (v) The general formula of the homologous series of alcohol is:

(a) CnH2n

(b) CnH2n+2

✓ (c) CnH2n+1OH

(d) CnH2n-1OH

Answer: (c) CnH2n+1OH — Alcohols are derived from alkanes (CnH2n+2) by replacing one hydrogen atom with the -OH group, giving the general formula CnH2n+1OH or R-OH. Examples are CH3OH (methanol) and C2H5OH (ethanol).

6. (vi) The number of five membered and six membered rings in C60 Bucky ball are respectively:

(a) 12 and 12

(b) 5 and 15

✓ (c) 12 and 20

(d) 40 and 20

Answer: (c) 12 and 20 — The C60 molecule (buckminsterfullerene) is a cage of 60 carbon atoms arranged like a football. It contains exactly 12 pentagonal (five-membered) rings and 20 hexagonal (six-membered) rings.

7. (vii) The pair of compounds that exhibit isomerism is:

(a) C2H5-OH and C3H7-OH

(c) C2H5-CH2-Cl and C3H7-CH2-Cl

(d) CH3-NH2 and CH3-CH2-NH2

Answer: (b) CH3-CH2-CH2-CH3 and CH3-CH(CH3)-CH3 — Both compounds have the same molecular formula C4H10 but different structures: one is n-butane and the other is isobutane (2-methylpropane), so they are structural isomers. In the other pairs the two compounds have different molecular formulas, so they are homologues, not isomers.

8. (viii) The type of coal that is hard and high ranked is:

(a) Peat

(b) Lignite

(c) Bituminous

✓ (d) Anthracite

Answer: (d) Anthracite — Anthracite is the highest-rank coal with the greatest carbon content (about 90% or more) and the highest calorific value. It is hard, shiny and burns with a clean, almost smokeless flame, unlike peat, lignite and bituminous coal which contain less carbon.

9. (ix) Which of the following hydrocarbon is the chief constituent of natural gas?

✓ (a) CH4

(b) C2H6

(c) C3H8

(d) C4H10

Answer: (a) CH4 — Natural gas consists mainly of methane, usually about 85-90%, with small amounts of ethane, propane and butane. Methane is therefore its chief constituent and the main source of its fuel value.

10. (x) Urea was first synthesized by Wohler from an inorganic material named as:

(a) Ammonium nitrate

(b) Ammonium chloride

✓ (c) Ammonium cyanate

(d) Ammonium bicarbonate

Answer: (c) Ammonium cyanate — In 1828 Friedrich Wohler heated ammonium cyanate, an inorganic compound, and obtained urea, an organic compound: NH4CNO -> CO(NH2)2. This synthesis disproved the vital force theory, which held that organic compounds could only be made by living organisms.

Short Questions

Question: 1. Define organic chemistry and give the significance of some organic compounds in everyday life

Answer: Organic chemistry is the branch of chemistry that deals with carbon compounds, mainly hydrocarbons and their derivatives. Organic compounds are essential in daily life: fuels such as petrol, LPG and natural

gas, medicines such as aspirin and antibiotics, plastics, synthetic fibres like nylon, soaps, detergents, dyes, perfumes and food items are all organic. Without them modern transport, medicine, clothing and industry would not be possible.

Question: 2. What is Vital force theory? Why was it disapproved?

Answer: The vital force theory stated that organic compounds could only be produced inside living organisms with the help of a mysterious vital force, and could never be prepared in the laboratory. It was disproved in 1828 when Wohler prepared urea, a typical organic compound, by heating the inorganic salt ammonium cyanate. After this, many organic compounds were synthesised in the laboratory, so the theory was abandoned.

Question: 3. Define functional group and write the structure of three oxygen containing functional group.

Answer: A functional group is an atom or a group of atoms in a molecule which is responsible for the characteristic chemical properties of that compound, for example -OH in alcohols. Three oxygen-containing functional groups are: hydroxyl group -OH (alcohol, R-OH); carbonyl group C=O in aldehydes as -CHO (R-CHO) and in ketones as R-CO-R; and carboxyl group -COOH (carboxylic acid, R-COOH). The ester group -COO- (R-COO-R) is another common oxygen-containing group.

Question: 4(a). Define the following terms: (a) Catenation

Answer: Catenation is the property of an element, especially carbon, to form long chains and rings by joining with atoms of the same element through covalent bonds. Carbon shows catenation to the maximum extent

because the C-C bond is strong and carbon is small and tetravalent. This property is a main reason for the existence of millions of organic compounds.

Question: 4(b). Define the following terms: (b) Isomerism

Answer: Isomerism is the phenomenon in which two or more compounds have the same molecular formula but different structures and therefore different properties; such compounds are called isomers. For example, n-butane and isobutane both have the formula C4H10 but different arrangements of carbon atoms. Isomerism may be structural, based on different connectivity, or stereo, based on different spatial arrangement.

Question: 5. How can you differentiate between total and partial synthesis of organic compounds?

Answer: In total synthesis an organic compound is built up in the laboratory starting from the elements themselves or from very simple inorganic materials, passing through all the steps in the laboratory. In partial synthesis the starting material is a natural organic compound obtained from plants or animals, which is then converted into the desired product by a few chemical steps. Total synthesis proves the structure of a compound completely, while partial synthesis is shorter and is often used commercially.

Question: 6. What is meant by Reforming? Why is it considered a useful technique?

Answer: Reforming is the process in which straight-chain alkanes of low-grade petrol are converted into branched-chain alkanes and aromatic hydrocarbons by heating with a catalyst such as platinum. It is useful

because branched and aromatic hydrocarbons have a high octane number and burn smoothly without knocking. Reforming therefore improves the quality of petrol for modern high-compression engines.

Question: 7. Write down some examples of products that can be produced using biotechnology?

Answer: Biotechnology produces useful materials using living organisms or enzymes. Examples include insulin for diabetic patients, antibiotics such as penicillin, vaccines, interferons and growth hormones. It is also used to produce ethanol fuel, enzymes for detergents, and improved seeds and crops in agriculture.

Question: 8. Write down some common uses of organic compounds?

Answer: Organic compounds are used as fuels (petrol, diesel, natural gas, coal), medicines (aspirin, paracetamol, antibiotics), and polymers (polythene, PVC, nylon, polyester) for plastics and fibres. They are also used as solvents, paints, dyes, perfumes, soaps, detergents, pesticides and fertilisers such as urea. In short, almost every industry from agriculture to medicine depends on organic compounds.

Descriptive Questions

Question: 1. Describe the natural sources of organic compounds.

Answer: The main natural sources are fossil remains, plants and animals. Coal, petroleum and natural gas were formed from the remains of plants and animals buried millions of years ago under high temperature and pressure, and they supply most hydrocarbons and fuels. Living plants provide carbohydrates, vegetable oils, alkaloids, rubber and wood, while animals provide fats, proteins, wool, silk, milk products and hormones.

These natural materials are the starting point for a very large number of organic products.

Question: 2. Define Bucky Ball? Explain its structure and mention its some properties.

Answer: Bucky ball is the common name of buckminsterfullerene, C60, an allotrope of carbon discovered in 1985. Its 60 carbon atoms form a hollow cage shaped like a football, made of 12 five-membered and 20 six-membered rings, with each carbon sp2 hybridised and part of a delocalised pi system. It is a black solid, insoluble in water but soluble in organic solvents such as benzene, and it can accept electrons to form salts. Fullerenes are used in research on superconductors, lubricants, drug delivery and new materials.

Question: 3. Explain destructive distillation of coal? What are the various products obtained from it?

Answer: Destructive distillation is the heating of coal strongly in the absence of air, so that it decomposes instead of burning. The products are: coke, the solid residue used as a fuel and reducing agent in metallurgy; coal tar, a thick black liquid from which benzene, toluene, phenol, naphthalene and many dyes and drugs are obtained; coal gas, a mixture of hydrogen, methane and carbon monoxide used as a fuel; and ammoniacal liquor, an aqueous solution of ammonia used to make fertilisers such as ammonium sulphate. It is an important industrial process carried out in coke ovens and gas works.

Question: 4. Describe homologous series of organic compounds? Write three main properties of homologous series.

Answer: A homologous series is a family of organic compounds having the same functional group and similar chemical properties, in which successive members differ by a -CH2- group, for example the alkanes CH4, C2H6, C3H8 and so on. Three main properties are: all members can be represented by one general formula, such as CnH2n+2 for alkanes; all members show similar chemical properties because they have the same functional group; and physical properties such as boiling point change gradually with increasing molecular mass. All members can also be prepared by similar general methods.

Question: 5. Explain various unique characteristics associated with organic compounds.

Answer: Organic compounds have several unique features. Carbon shows catenation and forms chains, branches and rings, and it forms strong covalent bonds with H, O, N, S and halogens, giving millions of compounds. They show isomerism, are generally covalent with low melting and boiling points, and are mostly insoluble in water but soluble in organic solvents. Their reactions are usually slow and molecular rather than ionic, and most of them are combustible and contain carbon that can be detected by burning.

Question: 6. Describe natural product chemistry and explain the use of plants and animals as sources of organic compounds.

Answer: Natural product chemistry deals with the isolation, structure and synthesis of organic compounds produced by living organisms. Plants are a rich source: they give carbohydrates, vegetable oils, essential oils, alkaloids such as quinine and morphine, natural rubber, and vitamins. Animals provide fats and oils, proteins, hormones such as insulin,

enzymes, wool and silk. Many important medicines were first obtained from these natural sources and are now also made in the laboratory.

Question: 7. Write down the different types of coal and how do they differ from each other?

Answer: Coal is classified by rank, that is, by increasing carbon content and calorific value, into peat, lignite, bituminous coal and anthracite. Peat is the first stage, soft and brown, with the least carbon (about 30-40% in dry form) and much moisture. Lignite (brown coal) contains more carbon and less moisture than peat. Bituminous coal is black, contains about 70-85% carbon and is the most widely used coal for industry and power generation. Anthracite is the highest rank, hard and shiny, with about 90% or more carbon, the highest heating value and a clean flame.

Chapter 4: Nomenclature of Organic Compounds

Multiple Choice Questions

1. (i) Which of the following molecule possess acyl functional group:

(a) R-CO-X

(b) R-CONH2

(c) R-COO-CH3

(d) R-CO-R

Answer: (a) R-CO-X — The acyl group is R-CO-, and in R-CO-X it is attached to a halogen atom, forming an acyl halide, the typical acyl compound. R-CO-R is a ketone, R-COO-CH3 is an ester, and the acyl group in an amide is present as part of the -CONH2 group rather than as the characteristic acyl halide group asked here.

2. (ii) A hydrocarbon with the molecular formula C7H12 is possibly:

(a) Heptane

(b) Heptene

(c) Heptyne

(d) Hepta diene

Answer: (c) Heptyne — C7H12 fits the alkyne general formula CnH2n-2 (for n = 7, 2n-2 = 12), so the hydrocarbon is heptyne with one triple bond. Heptane is C7H16 and heptene is C7H14, so they do not fit this formula.

3. (iii) An organic compound possesses the structural formula CH3CH = CH - C = CH (with a triple bond between the last two carbons, CH3-CH=CH-C triple bond CH), its correct IUPAC name is:

(a) 1-pentyne-3-ene

(b) 3-pentene-1-pentyne

(c) 3-penten-1-yne ✓

(d) 2-pentene-3-yne

Answer: (c) 3-penten-1-yne — The chain of five carbons is numbered from the end nearer the triple bond, giving the triple bond position 1 and the double bond position 3. The suffix is written as -en- before -yne, so the name is pent-3-en-1-yne, written in the book style as 3-penten-1-yne.

4. (iv) IUPAC name of isopropyl alcohol is:

(a) 1-propanol

(b) 2-propanol ✓

(c) 1-butanol

(d) 2-butanol

Answer: (b) 2-propanol — Isopropyl alcohol is (CH3)2CH-OH, in which the -OH group is attached to the middle carbon of a three-carbon chain. Numbering gives the -OH group position 2, so its IUPAC name is propan-2-ol or 2-propanol.

5. (v) Resorcinol is a phenol, it contains two -OH groups at:

(a) Position 1 and 2

(b) Position 1 and 3 ✓

(c) Position 1 and 4

(d) Position 2 and 4

Answer: (b) Position 1 and 3 — Resorcinol is benzene-1,3-diol, with the two -OH groups in meta positions to each other. The 1,2-diol is catechol and the 1,4-diol is hydroquinone (quinol).

6. (vi) Formula of a saturated hydrocarbon is C4H8, it should be:

(a) Butane

(b) Butene

(c) Butyne

✓ (d) Cyclobutane

Answer: (d) Cyclobutane — An open-chain saturated hydrocarbon would be C4H10 (butane), while butene C4H8 is unsaturated and butyne is C4H6. C4H8 can only be saturated if it is cyclic, because forming the ring removes two hydrogen atoms, so it is cyclobutane.

7. (vii) In the IUPAC naming of ketones the ending -e of main carbon chain is replaced by:

(a) yl

(b) ol

(c) al

✓ (d) one

Answer: (d) one — For ketones the final -e of the parent alkane is replaced by the suffix -one, as in propanone and butan-2-one. The suffix -ol is used for alcohols and -al for aldehydes.

8. (viii) In the common system, carboxylic acid with six carbon atoms in straight chain is named as:

(a) propionic acid

(b) valeric acid

✓ (c) caproic acid

(d) steric acid

Answer: (c) caproic acid — In the common system the straight-chain six-carbon acid CH3-(CH2)4-COOH is called caproic acid (IUPAC name hexanoic acid). Propionic acid has three carbons, valeric acid has five

carbons and stearic acid has eighteen carbons.

Question: 9. (ix) The correct structure of 1, 3-penta diene is: (a) H3C-CH = CH-HC = CH2-CH3 (b) CH2 = CH-CH2-CH = CH2 (c) CH2 = CH-HC = CH-CH3 (d) CH3-CH = C = CH-CH3

Answer: (c) CH2 = CH-HC = CH-CH3 — In 1,3-pentadiene the two double bonds start at carbons 1 and 3 of a five-carbon chain: CH2=CH-CH=CH-CH3. Option (b) is 1,4-pentadiene, and option (d) contains a cumulated (allene-type) arrangement, so neither represents 1,3-pentadiene.

10. (x) The IUPAC name of CH3COOCH(CH3)2 is:

(a) propyl ethanoate

(b) ethyl propanoate

(c) isopropyl acetate

✓ (d) isopropyl ethanoate

Answer: (d) isopropyl ethanoate — The acid part CH3COO- comes from ethanoic acid and the alcohol part -CH(CH3)2 is the isopropyl group. An ester is named as alkyl alkanoate, so the IUPAC name is isopropyl ethanoate; isopropyl acetate is only its common name.

Short Questions

Question: 1. Why IUPAC system of nomenclature is preferred on common system of naming of organic compounds?

Answer: The common (trivial) system gives names based on source or history, so one compound may have several names and the name tells nothing about its structure. The IUPAC system is systematic and universal: each compound gets one unique name from its structure, and from the

name the correct structure can be drawn anywhere in the world. It also works for the millions of complex compounds for which common names would be impossible to remember.

Question: 2. Write the name of five organic families and mention the functional group present in them.

Answer: Five organic families are: (1) Alcohols, functional group -OH (hydroxyl), for example ethanol. (2) Aldehydes, functional group -CHO, for example ethanal. (3) Ketones, functional group C=O (carbonyl) within the chain, for example propanone. (4) Carboxylic acids, functional group -COOH, for example ethanoic acid. (5) Amines, functional group -NH2, for example methylamine. Each family has its own suffix in IUPAC naming: -ol, -al, -one, -oic acid and -amine respectively.

Question: 3. Write down basic rules for IUPAC naming of organic compound.

Answer: The main rules are: select the longest continuous carbon chain containing the functional group as the parent chain; number the chain from the end that gives the functional group and then the substituents the lowest possible locants; name substituents as prefixes in alphabetical order, using di-, tri- for repeats; and write the name as prefixes + root word + primary suffix (ane, ene, yne) + secondary suffix for the functional group. Locants are separated by commas and are joined to the name by hyphens. For example, CH3-CH(CH3)-CH2-OH is 2-methylpropan-1-ol.

Question: 4. We often use the term Iso and Neo in the common system of naming of alkanes. Explain with example

Answer: In the common system, the prefix iso- is used when one methyl

group is attached to the second carbon of the chain, that is, the chain ends in a -CH(CH3)2 group; for example, (CH3)2CH-CH3 is isobutane and (CH3)2CH-CH2-OH is isobutyl alcohol. The prefix neo- is used when two methyl groups are attached to the second carbon, that is, the chain contains a -C(CH3)3 end group; for example, (CH3)4C is neopentane. These prefixes describe branching without using numbers.

Question: 5. Write the common names of first five members of carboxylic acid.

Answer: The first five members with their common names are: HCOOH formic acid (1 carbon), CH3COOH acetic acid (2 carbons), CH3CH2COOH propionic acid (3 carbons), CH3CH2CH2COOH butyric acid (4 carbons) and CH3(CH2)3COOH valeric acid (5 carbons). Their IUPAC names are methanoic, ethanoic, propanoic, butanoic and pentanoic acid respectively. The common names come from their natural sources, for example formic acid from ants and acetic acid from vinegar.

Question: 6. If an organic compound contains both double and triple bond, in the main carbon chain, what rules you follow to write its IUPAC name. Explain by giving an example.

Answer: The chain is numbered so that the multiple bonds together get the lowest possible locants; if there is still a choice, the double bond is given the lower locant. In the suffix, -ene is written before -yne, and the final -e of -ene is dropped, giving endings like -en-yne with both locants stated. For example, HC triple bond C-CH=CH2 is but-3-en-1-yne (numbered from the triple-bond end to give yne position 1), and CH3-CH=CH-C triple bond CH is pent-3-en-1-yne.

Descriptive Questions

Question: 1(a). Give the IUPAC names of the following organic molecules.

(a) CH3-C(=O)-O-CH(CH3)2

Answer: This is an ester formed from ethanoic acid and propan-2-ol, because the -O-CH(CH3)2 part is the isopropyl (propan-2-yl) group. Its IUPAC name is isopropyl ethanoate (propan-2-yl ethanoate). Esters are always named as alkyl alkanoate, the alkyl part being written first.

Question: 1(b). Give the IUPAC names of the following organic molecules.

(b) HC triple bond C-CH(CH3)-CH = CH2

Answer: The longest chain has five carbons and contains both a triple bond and a double bond, with a methyl branch on carbon 3. Numbering from the triple-bond end gives the triple bond position 1 and the double bond position 4. Its IUPAC name is 3-methylpent-4-en-1-yne.

Question: 1(c). Give the IUPAC names of the following organic molecules.

(c) CH2 = C(CH3)-COOH

Answer: The parent chain has three carbons including the -COOH carbon, which is carbon 1, and there is a double bond between carbons 2 and 3 with a methyl group on carbon 2. Its IUPAC name is 2-methylprop-2-enoic acid. Its common name is methacrylic acid, an important monomer for plastics.

Question: 1(d). Give the IUPAC names of the following organic molecules.

(d) (CH3)2CH-C(=O)-CH(Cl)-CH3, that is CH3-CH(CH3)-CO-CH(Cl)-CH3 drawn with a CH3 branch on the left CH and a Cl on the right CH

Answer: The longest chain containing the carbonyl group has five carbons, with the C=O group at carbon 3, a methyl group at carbon 2 and a

chlorine at carbon 4. Its IUPAC name is 4-chloro-2-methylpentan-3-one. Substituent prefixes are written in alphabetical order (chloro before methyl) with their locants.

Question: 1(e). Give the IUPAC names of the following organic molecules.

(e) CH3-CH2-O-C(CH3)2-C2H5

Answer: This is an ether in which oxygen joins an ethyl group and a C(CH3)2(C2H5) group. Treating it as a substituted butane, the longest chain is four carbons with an ethoxy group and a methyl group on carbon 2. Its IUPAC name is 2-ethoxy-2-methylbutane, also named ethyl (2-methylbutan-2-yl) ether in the functional-class system.

Question: 1(f). Give the IUPAC names of the following organic molecules.

(f) (CH3)2 CBr CHO

Answer: The -CHO group is carbon 1 of a three-carbon chain, and carbon 2 carries both a bromine atom and a methyl group. Its IUPAC name is 2-bromo-2-methylpropanal. Aldehydes take the suffix -al and the aldehyde carbon is always given position 1.

Question: 1(g). Give the IUPAC names of the following organic molecules.

(g) CH3-C(=O)-CH = CH-CH3

Answer: The chain has five carbons with the ketone group at carbon 2 and a double bond between carbons 3 and 4. Numbering is chosen to give the carbonyl group the lower locant. Its IUPAC name is pent-3-en-2-one.

Question: 1(h). Give the IUPAC names of the following organic molecules.

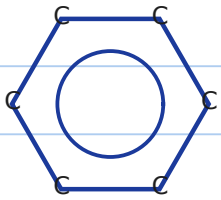
(h) CH2(OH)-CH(OH)-CH2(OH), glycerol structure with -OH on each carbon

Answer: This compound has a three-carbon chain with one hydroxyl group

on every carbon. Its IUPAC name is propane-1,2,3-triol. Its common name is glycerol (glycerine), obtained from fats and used in medicines, cosmetics and food.

Question: 1(i). Give the IUPAC names of the following organic molecules. (i) Benzene ring with -OH at the top and -C2H5 at the bottom (para positions)

Answer: The -OH group on the benzene ring makes the parent compound phenol, and the ethyl group is opposite to it, at position 4. Its IUPAC name is 4-ethylphenol. The ring is numbered from the carbon bearing the -OH group.



Six carbon atoms in a ring, one H on each carbon

Circle = delocalised pi electrons

All C-C bonds equal in length

Fig.: Benzene (labelled diagram)

Question: 1(j). Give the IUPAC names of the following organic molecules. (j) Benzene ring with -COCH3 and -Cl in meta positions

Answer: The -COCH3 (acetyl) group is taken as position 1 and the chlorine is two positions away from it, that is, at position 3 (meta). Its IUPAC name is 1-(3-chlorophenyl)ethan-1-one, commonly called 3-chloroacetophenone. Acetophenone itself is named systematically as 1-phenylethan-1-one.

Question: 2(a). Outline the structure of each of the following molecules. (a) Ethyl neo-pentyl ether

Answer: Neopentyl is the group (CH3)3C-CH2-, and ethyl is CH3CH2-, joined through an oxygen atom. The structure is: CH3CH2-O-CH2-C(CH3)3. It is an unsymmetrical ether because the two alkyl groups on oxygen are different.

Question: 2(b). Outline the structure of each of the following molecules. (b) Resorcinol

Answer: Resorcinol is benzene-1,3-diol: a benzene ring carrying two -OH groups at positions 1 and 3 (meta to each other). Its structure is written as C6H4(OH)2 with the OH groups separated by one ring carbon. It is used in antiseptics, dyes and resins.

Question: 2(c). Outline the structure of each of the following molecules. (c) Picric acid

Answer: Picric acid is 2,4,6-trinitrophenol: a phenol ring carrying three -NO2 groups at positions 2, 4 and 6 around the -OH group. Its structure is C6H2(NO2)3-OH. The three electron-withdrawing nitro groups make it a strong acid for a phenol, and it is also an explosive and a yellow dye.

Question: 2(d). Outline the structure of each of the following molecules. (d) Isopropylbutanoate

Answer: This ester is formed from butanoic acid and propan-2-ol, so the acid part is CH3-CH2-CH2-COO- and the alcohol part is -CH(CH3)2. Its structure is: CH3-CH2-CH2-C(=O)-O-CH(CH3)2. Like other low-molecular esters it has a fruity smell.

Question: 2(e). Outline the structure of each of the following molecules. (e) 1-Bromo-3-methyl hexane-2-one

Answer: The parent chain is six carbons with a ketone group at carbon 2, bromine at carbon 1 and a methyl branch at carbon 3. Its structure is: BrCH2-C(=O)-CH(CH3)-CH2-CH2-CH3. The carbonyl carbon gets the lowest possible locant after the chain is selected.

Question: 2(f). Outline the structure of each of the following molecules. (f) alpha, beta-dimethyl butyric acid

Answer: Butyric acid is a four-carbon acid; the carbon next to -COOH is alpha and the next is beta, and each carries one methyl group. Its structure is: CH3-CH(CH3)-CH(CH3)-COOH. In IUPAC terms the same compound is 2,3-dimethylbutanoic acid, since alpha = position 2 and beta = position 3.

Question: 2(g). Outline the structure of each of the following molecules. (g) Butyraldehyde

Answer: Butyraldehyde is the common name of the four-carbon aldehyde with a straight chain ending in -CHO. Its structure is: CH3-CH2-CH2-CHO. Its IUPAC name is butanal.

Question: 2(h). Outline the structure of each of the following molecules. (h) 3-Chlorobenzamide

Answer: Benzamide is a benzene ring attached to a -CONH2 group, which is taken as position 1; chlorine is placed at position 3 (meta). Its structure is written as Cl-C6H4-CONH2 with Cl meta to the amide group. Amides are the nitrogen derivatives of carboxylic acids.

Question: 2(i). Outline the structure of each of the following molecules. (i) Butanoyl iodide

Answer: Butanoyl iodide is the acyl halide of butanoic acid, with the -OH of the acid replaced by iodine. Its structure is: CH3-CH2-CH2-C(=O)-I. Acyl halides are highly reactive and fume in moist air.

Question: 2(j). Outline the structure of each of the following molecules. (j)

1, 2, 3-benzenetriol

Answer: This is a benzene ring carrying three -OH groups on three consecutive carbons, positions 1, 2 and 3. Its structure is C6H3(OH)3 with the three hydroxyl groups adjacent to one another. Its common name is pyrogallol, used in photography and hair dyes.

Question: 2(k). Outline the structure of each of the following molecules. (k) Ethyl isopropyl ketone

Answer: In the common ketone naming system the two groups on the carbonyl carbon are ethyl (CH3CH2-) and isopropyl ((CH3)2CH-). Its structure is: CH3-CH2-C(=O)-CH(CH3)2. Its IUPAC name is 2-methylpentan-3-one.

Question: 2(l). Outline the structure of each of the following molecules. (l) alpha-Methyl butyraldehyde

Answer: Butyraldehyde is CH3-CH2-CH2-CHO; the alpha carbon is the one next to the -CHO group, and it carries a methyl group. Its structure is: CH3-CH2-CH(CH3)-CHO. Its IUPAC name is 2-methylbutanal.

Question: 2(m). Outline the structure of each of the following molecules. (m) alpha-Chloro-beta-methyl valeric acid

Answer: Valeric acid is a five-carbon acid; chlorine is placed on the alpha carbon (position 2) and a methyl group on the beta carbon (position 3). Its structure is: CH3-CH2-CH(CH3)-CH(Cl)-COOH. In IUPAC terms it is 2-chloro-3-methylpentanoic acid.

Chapter 5: Hydrocarbons

Multiple Choice Questions

1. (i) The final product obtained when hydrogen bromide (HBr) is added to an ethyne molecule:

(a) Bromo ethene

(b) 1, 1-dibromo ethane ✓

(c) 1,2-bromo ethane

(d) 1,1,2,2 tetra bromoethane

Answer: (b) 1, 1-dibromo ethane — Ethyne adds two molecules of HBr in two steps, and both additions follow Markownikov rule, so both bromine atoms go to the same carbon. The final product is CH3-CHBr2, 1,1-dibromoethane; bromoethene is only the intermediate product of the first step.

2. (ii) The formula of a saturated hydrocarbon is C3H6, it should be:

(a) Propane

(b) Propene

(c) Propyne

(d) Cyclopropane ✓

Answer: (d) Cyclopropane — Propane is C3H8 and propene C3H6 is unsaturated, while propyne is C3H4. A saturated compound with formula C3H6 is only possible as a ring, cyclopropane, because ring formation removes two hydrogen atoms.

3. (iii) Ozonide on heating with zinc dust produce:

(a) Alcohol

(b) Aldehyde

(c) Alkene

(d) Ether

Answer: (b) Aldehyde — Reductive ozonolysis, in which the ozonide is decomposed with Zn and water, cleaves the double bond and gives carbonyl compounds, aldehydes being formed from =CH- groups. For example, ethene gives two molecules of formaldehyde (methanal), an aldehyde.

4. (iv) Which of the following pairs of compounds represent functional group isomerism?

(a) 1-butene and 2-butene

(b) Ethanol and dimethyl ether

(c) n-butane and iso butane

(d) Diethyl ketone and methyl propyl ketone

Answer: (b) Ethanol and dimethyl ether — Both have the molecular formula C2H6O but different functional groups: ethanol has the -OH (alcohol) group and dimethyl ether has the -O- (ether) group, so they are functional group isomers. The butenes show position isomerism and the butanes show chain isomerism.

5. (v) The substituent that can act as a meta director is:

(a) -Cl

(b) -CH3

(c) -OH

(d) -COOH

Answer: (d) -COOH — The carboxyl group is electron-withdrawing and deactivates the ring, directing the incoming electrophile to the meta

position. -Cl, -CH3 and -OH are all ortho/para directing groups.

6. (vi) Welding gas among the following is:

(a) Ethylene

✓ (b) Acetylene

(c) Ethane

(d) Methane

Answer: (b) Acetylene — Acetylene (ethyne) burns in oxygen with an extremely hot flame of about 3000 C, called the oxy-acetylene flame. This flame is hot enough to melt and join metals, so acetylene is used as the welding gas.

7. (vii) Benzene burns with smokey flame because of its:

(a) Inflammability

✓ (b) High carbon % age

(c) High resonance energy

(d) Aromaticity

Answer: (b) High carbon % age — Benzene, C6H6, contains about 92% carbon, and there is not enough air in an ordinary flame to burn all of it completely. The unburnt carbon particles glow and escape as soot, giving a smoky flame.

8. (viii) Select the suitable chemical to distinguish between Ethene and Ethyne:

(a) Alkaline KMnO4

(b) Acidified KMnO4

(c) Bromine water

✓ (d) Ammonical AgNO3

Answer: (d) Ammonical AgNO3 — Ethyne is a terminal alkyne with an acidic hydrogen and gives a white precipitate of silver acetylide with ammoniacal silver nitrate. Ethene gives no such precipitate, although both gases decolourise bromine water and KMnO4 solution.

9. (ix) Meta directing group among the following is:

(a) -OH

(b) -NH2

(c) -CH3

✓ (d) -NO2

Answer: (d) -NO2 — The nitro group is strongly electron-withdrawing, deactivates the benzene ring and directs electrophilic substitution to the meta position. -OH, -NH2 and -CH3 are activating, ortho/para directing groups.

10. (x) Acylation of benzene in the presence of AlCl3 gives:

(a) Toluene

✓ (b) Acetophenone

(c) Phenol

(d) Xylene

Answer: (b) Acetophenone — In Friedel-Crafts acylation, benzene reacts with acetyl chloride (CH3COCl) in the presence of anhydrous AlCl3 to give acetophenone, C6H5COCH3, a ketone. Reaction with CH3Cl under similar conditions (alkylation) would give toluene instead.

Short Questions

Question: 1. Give three differences between aliphatic and aromatic hydrocarbons?

Answer: (1) Aliphatic hydrocarbons have open-chain or non-aromatic ring structures, while aromatic hydrocarbons contain a benzene ring with delocalised pi electrons. (2) Aliphatic unsaturated compounds mainly give addition reactions, while aromatic compounds prefer electrophilic substitution reactions that preserve the ring. (3) Aromatic hydrocarbons have a much higher carbon percentage and burn with a smoky flame, whereas most aliphatic hydrocarbons burn with a comparatively clean flame.

Question: 2. Give the mechanism of free radical reaction between methane and chlorine in the presence of sunlight.

Answer: The reaction occurs in three stages. Initiation: sunlight splits a chlorine molecule into two free radicals, Cl2 -> 2Cl. (in the presence of hv). Propagation: Cl. + CH4 -> CH3. + HCl, and then CH3. + Cl2 -> CH3Cl + Cl., so the chain continues and more chlorine can be substituted to give CH2Cl2, CHCl3 and CCl4. Termination: two radicals combine and end the chain, for example Cl. + Cl. -> Cl2 and CH3. + CH3. -> C2H6.

Question: 3(a). Complete the following reactions and name the major product formed in each reaction. C2H5OH --H2SO4(conc.) 170 C-->

Answer: Ethanol is dehydrated by concentrated sulphuric acid at 170 C to give ethene. The completed reaction is: C2H5OH -> CH2=CH2 + H2O (conc. H2SO4, 170 C). The major product is ethene, formed by removal of -OH from one carbon and H from the adjacent carbon (elimination).

Question: 3(b). Complete the following reactions and name the major product formed in each reaction. C2H5Br + KOH --Alcohol, heat-->

Answer: With alcoholic KOH on heating, ethyl bromide undergoes

dehydrohalogenation (elimination of HBr) to give ethene. The completed reaction is: C2H5Br + KOH (alcoholic) -> CH2=CH2 + KBr + H2O. The major product is ethene; note that aqueous KOH would instead give ethanol by substitution.

Question: 3(c). Complete the following reactions and name the major product formed in each reaction. CH3-CH = CH2 + HBr -->

Answer: Propene adds HBr according to Markownikov rule: the hydrogen goes to the carbon already having more hydrogens, and bromine attaches to the middle carbon. The completed reaction is: CH3-CH=CH2 + HBr -> CH3-CHBr-CH3. The major product is 2-bromopropane.

Question: 3(d). Complete the following reactions and name the major product formed in each reaction. CH2 = CH2 + O3 -->

Answer: Ethene reacts with ozone to form an ozonide, which on decomposition with Zn/H2O gives two molecules of formaldehyde. The completed reaction is: CH2=CH2 + O3 -> (ethene ozonide) -> 2HCHO (with Zn/H2O). The major product is formaldehyde (methanal), and this reaction is called ozonolysis.

Question: 4. Define a chiral carbon? Give an example to justify your answer.

Answer: A chiral carbon is a carbon atom attached to four different atoms or groups, so that its molecule has no plane of symmetry and exists as non-superimposable mirror images (optical isomers). An example is the central carbon of lactic acid, CH3-CH(OH)-COOH, which is bonded to -H, -OH, -CH3 and -COOH, four different groups. Such a compound shows optical activity and rotates the plane of polarised light.

Question: 5(i). Write the equations with all required conditions for the reaction of benzene with the following: (i) H2

Answer: Benzene is hydrogenated in the presence of a nickel catalyst at about 150-200 C under pressure to give cyclohexane. The equation is: C6H6 + 3H2 -> C6H12 (Ni, heat, pressure). All three pi bonds of the ring are reduced in this addition reaction.

Question: 5(ii). Write the equations with all required conditions for the reaction of benzene with the following: (ii) CH3COCl

Answer: Benzene undergoes Friedel-Crafts acylation with acetyl chloride in the presence of anhydrous AlCl3 to give acetophenone. The equation is: C6H6 + CH3COCl -> C6H5COCH3 + HCl (anhydrous AlCl3). Acetophenone is an aromatic ketone with a pleasant smell.

Question: 5(iii). Write the equations with all required conditions for the reaction of benzene with the following: (iii) HNO3(conc.)

Answer: Benzene is nitrated by a mixture of concentrated HNO3 and concentrated H2SO4 (nitrating mixture) at about 50-60 C to give nitrobenzene. The equation is: C6H6 + HNO3 -> C6H5NO2 + H2O (conc. H2SO4, 50-60 C). Nitrobenzene is a pale yellow oily liquid smelling of bitter almonds.

Question: 5(iv). Write the equations with all required conditions for the reaction of benzene with the following: (iv) H2SO4(conc.)

Answer: Benzene is sulphonated by concentrated (fuming) sulphuric acid on heating to give benzene sulphonic acid. The equation is: C6H6 + H2SO4 -> C6H5SO3H + H2O (conc./fuming H2SO4, heat). The reaction is reversible and is an electrophilic substitution by the SO3 electrophile.

Question: 6(i). Bring about the following conversions. (i) Toluene to ortho-para nitro benzoic acid

Answer: First toluene is nitrated with conc. HNO3/conc. H2SO4 to give a mixture of o-nitrotoluene and p-nitrotoluene. The mixture is then oxidised with alkaline KMnO4 (or acidified K2Cr2O7) on heating, which converts the -CH3 group into -COOH. The overall conversion is: C6H5CH3 -> (nitration) O2N-C6H4-CH3 -> (oxidation, KMnO4/heat) O2N-C6H4-COOH, giving o-nitrobenzoic acid and p-nitrobenzoic acid, which are separated.

Question: 6(ii). Bring about the following conversions. (ii) Benzene to m-nitro toluene

Answer: The -NO2 (meta director) must be introduced before the -CH3 group. Step 1, nitration: C6H6 + conc. HNO3/conc. H2SO4 -> C6H5NO2 (nitrobenzene). Step 2, Friedel-Crafts alkylation: C6H5NO2 + CH3Cl/anhydrous AlCl3 -> m-CH3-C6H4-NO2 + HCl, because the nitro group directs the incoming methyl group to the meta position, giving m-nitrotoluene.

Question: 7. How can you prepare ethene from dehydration of ethanol and ethyne from dehydro halogenations of ethyl chloride?

Answer: Ethene is prepared by heating ethanol with excess concentrated H2SO4 at 170 C: C2H5OH -> CH2=CH2 + H2O; the acid acts as a dehydrating agent. For ethyne, ethyl chloride is first treated with alcoholic KOH to give ethene (CH3CH2Cl -> CH2=CH2), brominated to 1,2-dibromoethane (CH2Br-CH2Br), and then heated with strong alcoholic KOH (or NaNH2) which removes 2HBr: CH2Br-CH2Br -> HC triple bond CH + 2HBr. Thus double dehydrohalogenation of a vicinal dihalide prepared from ethyl chloride gives ethyne.

Question: 8. Alkanes are generally referred as paraffins due to their less reactivity, why are they stable towards chemical reactions?

Answer: The word paraffin means little affinity, and alkanes are unreactive because all their bonds are strong, non-polar C-C and C-H sigma bonds. They have no pi bonds, lone pairs or polar sites for reagents to attack, and their bond energies are high. They therefore do not react with acids, alkalis, oxidising agents or electrophiles under ordinary conditions, and react mainly by free-radical substitution in sunlight.

Descriptive Questions

Question: 1(i). Draw the orbital structures of the following hydrocarbons.
(i) Ethane

Answer: In ethane each carbon is sp3 hybridised and tetrahedral, with bond angles of about 109.5 degrees. The C-C bond is formed by head-on overlap of two sp3 orbitals (a sigma bond), and each C-H bond is formed by overlap of an sp3 orbital of carbon with the 1s orbital of hydrogen. The structure CH3-CH3 therefore contains seven sigma bonds and free rotation is possible about the C-C bond.

Question: 1(ii). Draw the orbital structures of the following hydrocarbons.
(ii) Ethylene

Answer: In ethylene (ethene) each carbon is sp2 hybridised and the molecule is planar with bond angles of about 120 degrees. The C-C sigma bond is formed by overlap of sp2 orbitals, the four C-H bonds by sp2-s overlap, and the second C-C bond is a pi bond formed by sideways overlap of the unhybridised p orbitals above and below the plane. Ethene thus has one sigma and one pi bond between the carbons, CH2=CH2, and rotation about the double bond is restricted.

Question: 1(iii). Draw the orbital structures of the following hydrocarbons. (iii) Acetylene

Answer: In acetylene (ethyne) each carbon is sp hybridised and the molecule is linear with a bond angle of 180 degrees. The C-C sigma bond comes from sp-sp overlap, the two C-H bonds from sp-s overlap, and two pi bonds are formed by sideways overlap of the two pairs of unhybridised p orbitals. Ethyne, HC triple bond CH, therefore contains one sigma and two pi bonds between the carbon atoms.

Question: 2(i). Write the equations of the following chemical processes. (i) Ethylene is heated at high temperature and pressure.

Answer: On heating under high pressure with a peroxide catalyst, ethene molecules join together to form the polymer polythene. The equation is: n(CH2=CH2) -> -(CH2-CH2)n- (high temperature, high pressure, catalyst). This is addition polymerisation, and polythene is used for bags, bottles and insulation.

Question: 2(ii). Write the equations of the following chemical processes. (ii) Ethene is burnt in air in the presence of per acetic acid.

Answer: In the presence of peracetic acid, ethene is oxidised (epoxidised) to ethylene oxide, a three-membered cyclic ether. The equation is: CH2=CH2 + CH3COOOH -> C2H4O (ethylene oxide, an epoxide ring) + CH3COOH. Ethylene oxide is an important industrial chemical used to make ethylene glycol and detergents.

Question: 2(iii). Write the equations of the following chemical processes. (iii) 1,2-di bromo butane is heated with alcoholic potassium hydroxide.

Answer: Strong alcoholic KOH removes two molecules of HBr from

1,2-dibromobutane (double dehydrohalogenation) to give but-1-yne. The equation is: CH3-CH2-CHBr-CH2Br + 2KOH (alc., heat) -> CH3-CH2-C triple bond CH + 2KBr + 2H2O. The product is 1-butyne, a terminal alkyne.

Question: 2(iv). Write the equations of the following chemical processes.
(iv) 1,1,2,2 tetra bromo ethane is heated with zinc powder.
Answer: Zinc removes bromine (debromination) from 1,1,2,2-tetrabromoethane on heating and a triple bond is formed, giving ethyne. The equation is: CHBr2-CHBr2 + 2Zn -> HC triple bond CH + 2ZnBr2 (heat). This is a laboratory method of preparing a pure alkyne from its tetrabromide.

Question: 2(v). Write the equations of the following chemical processes.
(v) Reaction of chlorine with acetylene
Answer: Acetylene adds chlorine across its triple bond in two steps, first giving 1,2-dichloroethene and then 1,1,2,2-tetrachloroethane. The equations are: HC triple bond CH + Cl2 -> CHCl=CHCl, and CHCl=CHCl + Cl2 -> CHCl2-CHCl2. The final product, 1,1,2,2-tetrachloroethane, is a heavy liquid formerly used as a solvent.

Question: 3. Ethene is more reactive than ethane but less reactive than ethyne, how can you explain this behavior?
Answer: Reactivity increases with the number of pi bonds, because pi electrons are loosely held, exposed and easily attacked by electrophiles, whereas the sigma bonds of ethane are strong and non-polar. Ethane has no pi bond and reacts only by slow free-radical substitution. Ethene has one pi bond and readily gives addition reactions, while ethyne has two pi bonds and an even greater density of exposed pi electrons, so it is the

most reactive of the three towards addition.

Question: 4. What is meant by isomerism? Explain four different types of structural isomers and give one example of each.

Answer: Isomerism is the existence of compounds with the same molecular formula but different structures and properties. Four types of structural isomerism are: (1) Chain isomerism, due to different carbon skeletons, for example n-butane and isobutane (C4H10). (2) Position isomerism, due to different positions of a functional group or multiple bond, for example 1-butene and 2-butene. (3) Functional group isomerism, due to different functional groups, for example ethanol and dimethyl ether (C2H6O). (4) Metamerism, due to different alkyl groups on either side of a polyvalent functional group, for example diethyl ether and methyl propyl ether (C4H10O).

Question: 5(a). Explain the following with suitable examples. (a) Optical isomers

Answer: Optical isomers (enantiomers) are stereoisomers which are non-superimposable mirror images of each other, arising when a molecule contains a chiral carbon attached to four different groups. They have identical physical and chemical properties except that they rotate plane-polarised light in opposite directions. An example is lactic acid, CH3-CH(OH)-COOH, which exists as (+) and (-) forms; a third example pair is the two forms of 2-butanol.

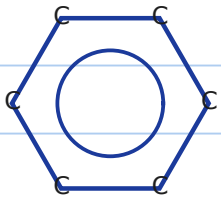
Question: 5(b). Explain the following with suitable examples. (b) Geometrical isomers

Answer: Geometrical (cis-trans) isomers are stereoisomers arising from

restricted rotation about a double bond, in which the groups attached to the double-bonded carbons are arranged differently in space. In cis-2-butene both -CH3 groups are on the same side of the double bond, while in trans-2-butene they are on opposite sides. The two isomers have different melting points, boiling points and dipole moments.

Question: 6. Describe the molecular orbital structure of benzene.

Answer: In benzene all six carbons are sp2 hybridised and form a planar hexagonal ring with bond angles of 120 degrees; the C-C sigma framework is made from sp2-sp2 overlaps and each C-H bond from sp2-s overlap. Each carbon has one unhybridised p orbital perpendicular to the ring, and the six p orbitals overlap sideways all around the ring to form a continuous delocalised pi cloud above and below the ring plane. This delocalisation makes all C-C bonds equal in length (between single and double bond values) and gives benzene its extra stability (resonance energy about 150 kJ/mol) and substitution rather than addition reactions.



Six carbon atoms in a ring, one H on each carbon

Circle = delocalised pi electrons

All C-C bonds equal in length

Fig.: Benzene (labelled diagram)

Question: 7(a). Give the mechanism of following Electrophilic substitution reaction of benzene. (a) Nitration

Answer: Step 1, formation of the electrophile: conc. HNO3 + conc. H2SO4 generate the nitronium ion, NO2+, the attacking electrophile (HNO3 + 2H2SO4 -> NO2+ + H3O+ + 2HSO4-). Step 2, the pi electrons of benzene attack NO2+ to form a resonance-stabilised carbocation (arenium ion) intermediate, in which aromaticity is temporarily lost. Step 3, HSO4- removes the proton from the carbon bearing the NO2 group, the pi system is restored and nitrobenzene is formed.

Question: 7(b). Give the mechanism of following Electrophilic substitution reaction of benzene. (b) Acylation

Answer: Step 1, acetyl chloride reacts with anhydrous AlCl3 to form the acylium ion electrophile, CH3-C+=O (CH3COCl + AlCl3 -> CH3CO+ + AlCl4-). Step 2, the benzene pi cloud attacks the acylium ion to give a resonance-stabilised arenium ion intermediate. Step 3, AlCl4- removes the ring proton, aromaticity is restored and acetophenone is formed, with AlCl3 regenerated at the end on aqueous work-up.

Question: 7(c). Give the mechanism of following Electrophilic substitution reaction of benzene. (c) Chlorination

Answer: Step 1, chlorine reacts with the Lewis acid catalyst FeCl3 (or AlCl3) to generate the electrophile Cl+ (Cl2 + FeCl3 -> Cl+ + FeCl4-). Step 2, benzene pi electrons attack Cl+ to form a resonance-stabilised arenium ion. Step 3, FeCl4- removes the proton from the same carbon, the aromatic pi system is restored, and chlorobenzene plus HCl are formed with the catalyst regenerated.

Question: 8. What is meant by ortho, para and meta directing groups. Explain the influence of substituent of benzene to the incoming Electrophile.

Answer: Positions 2 and 6 relative to a substituent are ortho, 3 and 5 are meta, and 4 is para. Activating groups with lone pairs or electron-donating effect, such as -OH, -NH2 and -CH3, increase electron density at the ortho and para positions, so they direct the incoming electrophile there and speed up substitution (o/p directors). Deactivating electron-withdrawing groups, such as -NO2, -COOH and -CHO, remove electron density most from the ortho and para positions, leaving the meta position relatively

richer, so substitution occurs at meta and more slowly (meta directors). Halogens are a special case: deactivating but still o/p directing.

Question: 9. Why ethyne terminal hydrogen is acidic in nature? Give two reactions of ethyne to show their acidic behavior.

Answer: In ethyne the carbon is sp hybridised and has 50% s-character, so it is strongly electronegative and holds the C-H electrons close, allowing the hydrogen to leave as H+ and leaving a stable acetylide ion. Two reactions showing this acidity are: (1) with sodium metal: 2HC triple bond CH + 2Na -> 2HC triple bond CNa (sodium acetylide) + H2; (2) with ammoniacal silver nitrate it gives a white precipitate: HC triple bond CH + 2[Ag(NH3)2]OH -> AgC triple bond CAg (silver acetylide) + 4NH3 + 2H2O. Internal alkynes and alkenes do not show these reactions.

Chapter 6: Alkyl Halides and Amines

Multiple Choice Questions

1. (i) Which of the following composition justifies the secondary alkyl halide?

(a) R3CX

✓ (b) R2CHX

(c) RCH2X

(d) CH3X

Answer: (b) R2CHX — In a secondary alkyl halide the halogen-bearing carbon is attached to two other carbon atoms, which is exactly the composition R2CHX. RCH2X is primary, R3CX is tertiary, and CH3X is a methyl halide.

2. (ii) Which of the following alkyl halide cannot produce an alkene while treated with alcoholic potassium hydroxide:

✓ (a) Methyl bromide

(b) Ethyl bromide

(c) Propyl bromide

(d) Butyl bromide

Answer: (a) Methyl bromide — Alkene formation by dehydrohalogenation needs hydrogen on a carbon adjacent to the C-Br carbon (a beta-hydrogen). Methyl bromide, CH3Br, has only one carbon and no beta-carbon at all, so it cannot form an alkene with alcoholic KOH.

3. (iii) Ethyl magnesium bromide with carbon dioxide yields.

(a) Methanoic acid

(b) Ethanoic acid

✓ (c) Propanoic acid

(d) Butanoic acid

Answer: (c) Propanoic acid — The Grignard reagent adds to CO2 and hydrolysis of the adduct gives a carboxylic acid with one carbon more than the alkyl group. Ethyl (2 carbons) plus the CO2 carbon gives the 3-carbon acid: C2H5MgBr + CO2 -> C2H5COOH, propanoic acid.

4. (iv) Grignard reagent with ester produces:

(a) Aldehyde

(b) Carboxylic acid

✓ (c) Ketone

(d) Ether

Answer: (c) Ketone — A Grignard reagent first displaces the alkoxy group of an ester to give a ketone: RCOOR + R-MgX -> R-CO-R (with RMgX and work-up). With a second equivalent of Grignard reagent the ketone reacts further to give a tertiary alcohol, but the ketone is the product of the first stage asked here.

5. (v) Amines act as bases because:

(a) They accept OH-

✓ (b) They accept H+

(c) They donate H+

(d) They donate the OH-

Answer: (b) They accept H+ — The nitrogen atom of an amine has a lone pair of electrons which it donates to a proton, forming an alkylammonium ion: R-NH2 + H+ -> R-NH3+. Accepting a proton is the behaviour of a base, so amines are basic in nature.

6. (vi) The structure of Primary amine is:

(a) Planar trigonal

(b) Linear

✓ (c) Tetrahedral pyramidal

(d) Regular tetrahedral

Answer: (c) Tetrahedral pyramidal — In a primary amine the nitrogen is bonded to three atoms (one C and two H) and carries one lone pair, giving a pyramidal arrangement based on a tetrahedron, similar to ammonia. The bond angle is about 107 degrees, slightly less than the tetrahedral angle because the lone pair repels the bond pairs.

7. (vii) Alkyl amine when reacts with nitrous acid in the presence of hydrochloric acid, yields:

(a) Diazonium salt

(b) Aldehyde

(c) Ketone

✓ (d) Alcohol

Answer: (d) Alcohol — A primary aliphatic (alkyl) amine reacts with nitrous acid ($NaNO_2$ + HCl) to form an unstable diazonium salt which immediately decomposes, giving the corresponding alcohol with evolution of nitrogen: $R\text{-}NH_2 + HNO_2 \rightarrow R\text{-}OH + N_2 + H_2O$. Stable diazonium salts are formed only by aromatic primary amines at low temperature.

8. (viii) SN2 reaction occurs most easily if the substrate molecule is:

✓ (a) A methyl iodide

(b) An ethyl iodide

(c) 2-iodo propane

(d) 2-iodo butane

Answer: (a) A methyl iodide — SN2 attack takes place from the back side of the C-X bond, so the reaction is fastest when there is least steric hindrance around the carbon. Methyl iodide is the least hindered of the given substrates, so it undergoes SN2 most easily; the order is methyl > primary > secondary.

9. (ix) Suitable reagent required for the synthesis of propane from methyl magnesium iodide is:

✓ (a) H2O

(b) NH3

(c) CH3OH

(d) CH3NH2

Answer: (a) H2O — Water is the standard reagent used to decompose (hydrolyse) a Grignard reagent: the alkyl group takes a proton from water and the corresponding alkane is formed, RMgX + H2O -> R-H + Mg(OH)X. Note on accuracy: the alkane formed always has the same carbon skeleton as the Grignard alkyl group, so methyl magnesium iodide itself gives methane, CH3MgI + H2O -> CH4 + Mg(OH)I; the book intends this hydrolysis step, and propane is obtained by the same method from propyl magnesium halide, C3H7MgX + H2O -> C3H8.

10. (x) The rate of SN1 mechanism depends upon:

✓ (a) Conc. of substrate only

(b) Conc. of attacking nucleophile only

(c) Conc. of both substrate and attacking nucleophile

(d) Polar solvent

Answer: (a) Conc. of substrate only — In SN1 the slow, rate-determining step is the ionisation of the alkyl halide alone to form the carbocation, in

which the nucleophile plays no part. The rate law is therefore rate = k[RX], first order in substrate only. SN2, by contrast, depends on the concentration of both reactants.

Short Questions

Question: 1(i). How are alkyl halides prepared by the reaction of alcohol with (i) HX (ii) PX3 (iii) SOCl2. Give the equations. (i) HX

Answer: Alcohols react with hydrogen halides to form alkyl halides, the -OH group being replaced by the halogen (ZnCl2 is used as catalyst with HCl). The equation is: R-OH + HX -> R-X + H2O, for example C2H5OH + HBr -> C2H5Br + H2O. The reactivity order of the acids is HI > HBr > HCl, and tertiary alcohols react fastest.

Question: 1(ii). How are alkyl halides prepared by the reaction of alcohol with (i) HX (ii) PX3 (iii) SOCl2. Give the equations. (ii) PX3

Answer: Phosphorus trihalides convert alcohols into alkyl halides, three molecules of alcohol reacting with one molecule of PX3. The equation is: 3R-OH + PX3 -> 3R-X + H3PO3, for example 3C2H5OH + PBr3 -> 3C2H5Br + H3PO3. This method gives good yields, especially for bromides and iodides prepared in situ from red phosphorus and the halogen.

Question: 1(iii). How are alkyl halides prepared by the reaction of alcohol with (i) HX (ii) PX3 (iii) SOCl2. Give the equations. (iii) SOCl2

Answer: Thionyl chloride converts an alcohol into the alkyl chloride with the gaseous by-products SO2 and HCl, so the halide is obtained pure. The equation is: R-OH + SOCl2 -> R-Cl + SO2 + HCl, for example C2H5OH + SOCl2 -> C2H5Cl + SO2 + HCl. A little pyridine is often added to absorb the HCl formed.

Question: 2. How are amines prepared from nitriles, give the equations.

Answer: Nitriles (alkyl cyanides) are reduced with hydrogen in the presence of a nickel catalyst, or with LiAlH4 or sodium and alcohol, to give primary amines having one carbon more than the original alkyl halide used to make the nitrile. The equation is: R-C triple bond N + 4[H] -> R-CH2-NH2, for example CH3-CN + 4[H] -> CH3-CH2-NH2 (ethylamine). This is a useful method because the chain is lengthened by one carbon atom.

Question: 3. Why are secondary and tertiary amines more alkaline than primary amines?

Answer: Alkyl groups are electron-donating (+I effect), so they push electron density onto the nitrogen and make its lone pair more available for accepting a proton; they also stabilise the alkylammonium ion formed. A secondary amine has two alkyl groups and a tertiary amine has three, so the effect is greater than in a primary amine with only one alkyl group. In aqueous solution the order is modified by solvation and steric effects, but the inductive effect explains why substituted amines are stronger bases than ammonia and, in the gas phase, tertiary > secondary > primary.

Question: 4. Why the alkyl part of Grignard reagent is nucleophilic in nature?

Answer: In a Grignard reagent, RMgX, magnesium is much less electronegative than carbon, so the C-Mg bond is polarised as C delta- and Mg delta+. The alkyl group therefore carries a partial negative charge and a pair of electrons it can donate, behaving like a carbanion. It attacks electron-deficient (electrophilic) centres such as the carbonyl carbon, so the alkyl part acts as a nucleophile and a strong base.

Question: 5(a). Convert the followings: (a) Methyl magnesium bromide into acetone

Answer: Step 1: methyl magnesium bromide is treated with carbon dioxide and the adduct is hydrolysed to give ethanoic acid: CH3MgBr + CO2 -> CH3COOH (after hydrolysis). Step 2: the acid is neutralised with Ca(OH)2 to give calcium acetate, (CH3COO)2Ca. Step 3: dry distillation of calcium acetate gives acetone: (CH3COO)2Ca -> CH3-CO-CH3 + CaCO3 on strong heating. Thus CH3MgBr is converted into acetone through ethanoic acid and its calcium salt.

Question: 5(b). Convert the followings: (b) Ethylchloride into ethyl amine

Answer: Ethyl chloride is heated with an excess of alcoholic ammonia under pressure (ammonolysis), when the chlorine is replaced by the -NH2 group. The equation is: C2H5Cl + NH3 (alc., excess, heat) -> C2H5NH2 + HCl, the amine being obtained as its salt and liberated by alkali. Excess ammonia is used to reduce further alkylation to secondary and tertiary amines.

Question: 5(c). Convert the followings: (c) Ethyl chloride into ethyl alcohol

Answer: Ethyl chloride is hydrolysed by boiling with aqueous potassium hydroxide (or moist silver oxide), the chlorine being replaced by the -OH group. The equation is: C2H5Cl + KOH (aq.) -> C2H5OH + KCl. This is a nucleophilic substitution reaction by the OH- ion; alcoholic KOH must be avoided because it gives ethene instead.

Question: 5(d). Convert the followings: (d) Ethylamine into imine

Answer: Ethylamine is dehydrogenated (oxidised) to the imine ethanimine, in which the -NH2 group becomes a C=N double bond: CH3-CH2-NH2 ->

CH3-CH=NH + H2 (oxidation, removal of hydrogen). The product CH3-CH=NH contains the characteristic imine group C=N-H. Imine formation by condensation of an amine with an aldehyde or ketone with loss of water gives substituted imines (Schiff bases) in the same family.

Question: 6. How is primary amine converted into secondary and tertiary amines, give the equations.

Answer: A primary amine is alkylated by heating it with an alkyl halide; the nitrogen attacks the alkyl halide and takes a second alkyl group to become a secondary amine, and on further alkylation a tertiary amine and finally a quaternary ammonium salt are formed. The equations are: R-NH2 + R-X -> R2NH + HX (secondary amine), and R2NH + R-X -> R3N + HX (tertiary amine). With excess RX the tertiary amine gives the quaternary salt [R4N]+ X-.

Question: 7. Give an account on the basicity of amines.

Answer: Amines are basic because the nitrogen lone pair can accept a proton from water or an acid: R-NH2 + H2O <-> R-NH3+ + OH-. Alkyl groups increase basicity by their +I effect, so aliphatic amines are stronger bases than ammonia. Aromatic amines such as aniline are much weaker bases, because the nitrogen lone pair is delocalised into the benzene ring and is less available for protonation. Basic strength is expressed by Kb or by the pKa of the conjugate acid.

Question: 8. How can you justify the fact that alkyl halides are water insoluble?

Answer: Alkyl halides are polar molecules, but they cannot form hydrogen bonds with water because they have no hydrogen attached to a strongly

electronegative atom. The energy released when halide molecules are surrounded by water is not enough to break the strong hydrogen bonds already present between water molecules. They therefore remain almost insoluble in water, although they dissolve readily in organic solvents of similar nature.

Question: 9. How can you define a nucleophile? Write the names of four nucleophiles along with their typical reagents.

Answer: A nucleophile is an electron-pair donor, a species with a lone pair or negative charge that attacks an electron-deficient carbon centre and forms a new covalent bond. Four examples with their typical reagents are: hydroxide ion OH- (aqueous KOH or NaOH), cyanide ion CN- (KCN in alcohol), ammonia NH3 (alcoholic ammonia), and alkoxide ion RO- (sodium alkoxide, RONa). Halide ions such as I- (KI) are also common nucleophiles.

Question: 10. Why the mechanism of SN2 reaction completes in one step?

Answer: In SN2 the nucleophile attacks the carbon from the back side at the same time as the leaving group departs, passing through a single transition state in which the carbon is partially bonded to both groups. Because bond-making and bond-breaking are simultaneous and no carbocation intermediate is formed, the whole substitution is completed in one concerted step. This also explains the inversion of configuration (Walden inversion) observed in the product.

Descriptive Questions

Question: 1. How can you define nucleophilic substitution reactions? Describe the mechanisms of SN1 and SN2 reactions.

Answer: A nucleophilic substitution is a reaction in which a nucleophile

replaces an atom or group (the leaving group) attached to a saturated carbon, for example R-X + OH- -> R-OH + X-. In the SN1 mechanism the reaction occurs in two steps: the halide slowly ionises to a planar carbocation and X-, and the nucleophile then attacks the carbocation rapidly from either side, giving a racemic product; the rate depends only on [RX] and tertiary halides react fastest. In the SN2 mechanism the nucleophile attacks from the back side in one concerted step through a transition state, the product shows complete inversion of configuration, the rate depends on [RX] and [Nu-], and methyl and primary halides react fastest.

Question: 2. What is beta-elimination? Discuss the mechanisms of E1 and E2 reactions.

Answer: Beta-elimination is the removal of a hydrogen atom from the beta-carbon (the carbon adjacent to the one bearing the leaving group) together with the leaving group itself, forming a double bond; dehydrohalogenation of alkyl halides with alcoholic KOH is the common example. In the E1 mechanism the halide first ionises slowly to a carbocation, and a base then removes a beta-proton to give the alkene, so the rate depends only on [RX]. In the E2 mechanism a strong base removes the beta-proton at the same time as the halide leaves, in a single concerted step through one transition state, so the rate depends on both [RX] and [base]. In both cases the more substituted alkene (Saytzeff product) usually predominates.

Question: 3. What are organometallic compounds? How is Grignard reagent prepared? Write down the reactions of Grignard reagent with water, carbon dioxide, ester and amines.

Answer: Organometallic compounds are organic compounds containing a direct carbon-metal bond, for example CH3MgBr and (C2H5)4Pb. A Grignard reagent is prepared by adding an alkyl halide to dry magnesium turnings in dry ether: R-X + Mg -> R-MgX (dry ether), the ether and apparatus being kept completely free of moisture. Its reactions are: with water it gives the alkane, RMgX + H2O -> R-H + Mg(OH)X; with carbon dioxide followed by hydrolysis it gives a carboxylic acid with one extra carbon, RMgX + CO2 -> R-COOH; with an ester it gives first a ketone and with excess reagent a tertiary alcohol; and with an amine (or any compound with an acidic hydrogen) it is simply protonated to the alkane, RMgX + H2N-R -> R-H + BrMg-NH-R.

Question: 4. What are alkyl halides? Define primary, secondary and tertiary alkyl halides.

Answer: Alkyl halides (haloalkanes) are organic compounds in which one or more hydrogen atoms of an alkane have been replaced by halogen atoms, general formula CnH2n+1X, for example C2H5Br. They are classified by the carbon atom carrying the halogen: in a primary (1 degree) alkyl halide the halogen-bearing carbon is attached to only one other carbon, for example 1-bromopropane, CH3CH2CH2Br. In a secondary (2 degree) alkyl halide it is attached to two other carbons, for example 2-bromopropane, CH3-CHBr-CH3, and in a tertiary (3 degree) alkyl halide it is attached to three other carbons, for example 2-bromo-2-methylpropane, (CH3)3CBr. Reactivity towards SN1 increases and towards SN2 decreases in the order primary < secondary < tertiary reversed respectively.

Question: 5. Draw the orbital structure of methyl iodide and explain the type of hybridization in it.

Answer: In methyl iodide, CH3I, the carbon atom is sp3 hybridised: its one 2s and three 2p orbitals mix to form four sp3 hybrid orbitals arranged tetrahedrally with angles of about 109.5 degrees. Three sp3 orbitals overlap with the 1s orbitals of the three hydrogens to form three C-H sigma bonds. The fourth sp3 orbital overlaps with a p orbital of iodine to form the polar C-I sigma bond, which is longer and weaker than a C-H bond. The molecule is therefore tetrahedral, and the polarity of the C-I bond makes CH3I highly reactive in nucleophilic substitution.

Question: 6. Give a comparative study between nucleophilic substitution reactions and elimination reactions of alkyl halides.

Answer: In substitution a nucleophile replaces the halogen and the carbon remains saturated, for example C2H5Br + KOH (aq.) -> C2H5OH + KBr, while in elimination the halogen and a beta-hydrogen are removed and an unsaturated product is formed, C2H5Br + KOH (alc.) -> C2H4 + KBr + H2O. Substitution is favoured by aqueous alkali, lower temperature and a good nucleophile that is a weak base, whereas elimination is favoured by alcoholic alkali, higher temperature and a strong bulky base. Both reactions compete for the same substrate, substitution dominating for primary halides and elimination becoming more important for secondary and tertiary halides under basic conditions. Mechanistically, SN1 competes with E1 through a common carbocation, and SN2 competes with E2 as concerted pathways.

Chapter 7: Alcohols, Phenols and Ethers

Multiple Choice Questions

1. (i) In the molecule of phenol, the carbon atom which is attached to hydroxyl group is.

(a) sp-hybridized

✓ (b) sp2-hybridized

(c) sp3-hybridized

(d) Unhybridized

Answer: (b) sp2-hybridized — All carbon atoms of the benzene ring in phenol, including the one bearing the -OH group, are sp2 hybridised and part of the flat aromatic ring. This allows the oxygen lone pair to conjugate with the ring pi system.

2. (ii) Which of the following is a trihydric phenol?

(a) Resorcinol

(b) Cresol

✓ (c) Pyrogallol

(d) Catechol

Answer: (c) Pyrogallol — Pyrogallol is benzene-1,2,3-triol and carries three -OH groups on the ring, so it is a trihydric phenol. Resorcinol and catechol are dihydric (two -OH groups) and cresol is monohydric.

3. (iii) Ethanol reacts with PCl3 to form.

(a) Diethyl ether

(b) Ethene

✓ (c) Ethyl chloride

(d) Ethanoic acid

Answer: (c) Ethyl chloride — Phosphorus trichloride replaces the -OH group of the alcohol by chlorine: 3C2H5OH + PCl3 -> 3C2H5Cl + H3PO3. Ethyl chloride, a gas used as a local anaesthetic spray, is therefore the product.

4. (iv) Which of the following alcohols has highest boiling Point

(a) Ethyl alcohol

✓ (b) n-pentyl alcohol

(c) iso-pentyl alcohol

(d) neo-pentyl alcohol

Answer: (b) n-pentyl alcohol — Boiling point increases with molecular mass and with chain straightness, because a straight chain has a larger surface area and stronger van der Waals forces. n-Pentyl alcohol (pentan-1-ol, b.p. about 138 C) therefore boils higher than its branched isomers isopentyl and neopentyl alcohol, and much higher than ethyl alcohol.

5. (v) Which of the following Products is mainly formed if ethanol is dehydrated with concentrated sulphuric acid at 170 C?

✓ (a) Ethene

(b) Ethyne

(c) Ethanol

(d) Diethyl ether

Answer: (a) Ethene — At 170 C concentrated H2SO4 removes a molecule of water from ethanol intramolecularly: C2H5OH -> CH2=CH2 + H2O. At the lower temperature of about 140 C the main product would instead be diethyl ether.

6. (vi) Lucas reagent is a mixture of

(a) Zn and Hg

(b) Zn and HCl

✓ (c) ZnCl2 and HCl

(d) NaOH and CaO

Answer: (c) ZnCl2 and HCl — Lucas reagent is a solution of anhydrous zinc chloride in concentrated hydrochloric acid. It is used to distinguish primary, secondary and tertiary alcohols by the speed with which turbidity (alkyl chloride) appears.

7. (vii) Oxidative cleavage of 1, 2-diol occur in the presence of

(a) K2Cr2O7

(b) KMnO4

(c) HNO3

✓ (d) HIO4

Answer: (d) HIO4 — Periodic acid, HIO4, specifically cleaves the C-C bond of a vicinal (1,2-) diol, each -CH(OH)- group being converted into a carbonyl group. For example, ethylene glycol gives two molecules of formaldehyde: HOCH2-CH2OH + HIO4 -> 2HCHO + HIO3 + H2O.

8. (viii) Which of the following molecule cannot form hydrogen bonding with water molecule?

(a) Phenol

(b) Resorcinol

✓ (c) Ethyl chloride

(d) Ethyl alcohol

Answer: (c) Ethyl chloride — Phenol, resorcinol and ethyl alcohol all contain -OH groups and form hydrogen bonds with water, which makes them appreciably soluble. Ethyl chloride has no -OH group and cannot

hydrogen-bond with water, so it is practically insoluble in water.

9. (ix) Secondary alcohols, undergo oxidation with potassium dichromate to produce carboxylic acid through an intermediate product known as:

(a) Aldehyde

✓ (b) Ketone

(c) Ether

(d) Alkyl halide

Answer: (b) Ketone — A secondary alcohol is first oxidised to a ketone, R2CH-OH -> R2C=O, and only under vigorous conditions is the ketone further cleaved to carboxylic acids with fewer carbon atoms. The ketone is therefore the intermediate product of the oxidation, unlike primary alcohols which pass through an aldehyde.

10. (x) Which of the following is an anaesthetic agent

(a) Phenol

(b) Ethyl alcohol

✓ (c) Diethyl ether

(d) Acetone

Answer: (c) Diethyl ether — Diethyl ether was the first widely used general anaesthetic and produces unconsciousness when its vapour is inhaled. It is highly volatile and inflammable, which is why safer anaesthetics have largely replaced it in modern surgery.

Short Questions

Question: 1. Define Phenol? Write the equations for the preparation of Phenol from. (i) Chlorobenzene (ii) Sodium benzene sulphonate

Answer: Phenol, C6H5OH, is the aromatic compound in which a hydroxyl

group is attached directly to a benzene ring; it is also called carbolic acid because it is weakly acidic. (i) From chlorobenzene (Dow process): chlorobenzene is fused with aqueous NaOH at about 350 C and 150 atmospheres pressure to give sodium phenoxide, which on acidification gives phenol: C6H5Cl + 2NaOH -> C6H5ONa + NaCl + H2O, then C6H5ONa + HCl -> C6H5OH + NaCl. (ii) From sodium benzene sulphonate: the salt is fused with NaOH at about 300 C to give sodium phenoxide and sodium sulphite: C6H5SO3Na + 2NaOH -> C6H5ONa + Na2SO3 + H2O, and the phenoxide is then acidified to phenol as above.

Question: 2(i). Write the equations for the following chemical Process (i) Reduction of acetic acid with LiAlH4.

Answer: Lithium aluminium hydride is a powerful reducing agent and reduces acetic acid all the way to the primary alcohol ethanol. The equation is: CH3COOH -> CH3CH2OH (LiAlH4, dry ether, then H2O work-up). The -COOH group is converted into -CH2OH, and water alone cannot be used as solvent because LiAlH4 reacts violently with it.

Question: 2(ii). Write the equations for the following chemical Process (ii) Hydration of ethene with hot concentrated H2SO4

Answer: Ethene is absorbed in hot concentrated sulphuric acid to form ethyl hydrogen sulphate, which on hydrolysis with boiling water gives ethanol. The equations are: CH2=CH2 + H2SO4 -> CH3CH2OSO3H, and CH3CH2OSO3H + H2O -> C2H5OH + H2SO4 (boil). This is the indirect hydration method formerly used for industrial ethanol.

Question: 2(iii). Write the equations for the following chemical Process (iii) Oxidation of ethanol with acidified dichromate.

Answer: With acidified potassium dichromate, ethanol is oxidised first to ethanal and then to ethanoic (acetic) acid, the orange dichromate turning green. The overall equation is: CH3CH2OH + 2[O] -> CH3COOH + H2O (K2Cr2O7/H2SO4). If the ethanal is distilled off as it forms, it can be isolated as the product instead.

Question: 2(iv). Write the equations for the following chemical Process (iv) Hydrolysis of diazonium salt

Answer: A benzene diazonium salt solution, prepared by diazotising aniline at 0-5 C, on warming with water hydrolyses to phenol with evolution of nitrogen. The equation is: C6H5N2+ Cl- + H2O -> C6H5OH + N2 + HCl (warm). This is an important laboratory preparation of phenol.

Question: 3(i). Explain the following with scientific reason (i) Boiling point of ether is less than alcohol?

Answer: Alcohol molecules are associated by strong intermolecular hydrogen bonding through their -OH groups, so much heat is needed to separate them. Ether molecules have no hydrogen attached to oxygen and cannot form hydrogen bonds among themselves; only weak dipole forces hold them together. Ethers therefore boil at much lower temperatures than alcohols of comparable molecular mass.

Question: 3(ii). Explain the following with scientific reason (ii) Alcohols are soluble in water?

Answer: The -OH group of an alcohol can form hydrogen bonds with water molecules, and this attraction compensates for breaking the hydrogen bonds in water itself. Lower alcohols such as methanol and ethanol are therefore miscible with water in all proportions. Solubility decreases as the

non-polar alkyl chain becomes longer, because the hydrocarbon part opposes dissolution.

Question: 3(iii). Explain the following with scientific reason (iii) Ethanol is liquid but ethyl chloride is gas at room temperature?

Answer: Ethanol molecules are held together by strong intermolecular hydrogen bonds, so ethanol is associated into larger clusters and is a liquid (b.p. 78 C). Ethyl chloride molecules have no hydrogen bonding and only weak van der Waals forces, so they escape easily as a gas (b.p. about 12 C). The difference in state is thus due to hydrogen bonding in ethanol.

Question: 4(i). Identify each of following with two laboratory tests. (i) Phenol

Answer: Phenol can be identified by two tests. (1) Ferric chloride test: phenol gives a violet (purple) colouration with neutral FeCl3 solution. (2) Bromine water test: phenol decolourises bromine water and gives a white precipitate of 2,4,6-tribromophenol. It also turns blue litmus slightly red, showing its weak acidity.

Question: 4(ii). Identify each of following with two laboratory tests. (ii) Alcohol

Answer: An alcohol can be identified by two tests. (1) Sodium metal test: an alcohol gives brisk effervescence of hydrogen with a small piece of sodium, 2R-OH + 2Na -> 2R-ONa + H2. (2) Ester test: warmed with acetic acid and a few drops of concentrated H2SO4 it gives a sweet, fruity smell of the ester formed. Ethanol also gives the iodoform test, a yellow precipitate of CHI3 with I2/NaOH.

Question: 5. What is Lucas reagent? Describe its use to distinguish between primary, secondary and tertiary alcohol.

Answer: Lucas reagent is a mixture of anhydrous ZnCl2 and concentrated HCl, which converts alcohols into insoluble alkyl chlorides that make the solution turbid. A tertiary alcohol reacts immediately and the mixture becomes turbid at once. A secondary alcohol becomes turbid within about five minutes. A primary alcohol does not react at room temperature and the mixture stays clear, reacting only on heating. The test works because tertiary carbocations form most easily.

Question: 6. What is oxonium ion? How can ether form oxonium ion?

Answer: An oxonium ion is a positively charged oxygen species in which oxygen carries three bonds and one lone pair, for example H3O+ and R2OH+. An ether oxygen has two lone pairs; in strong acid it accepts a proton on one lone pair to form a dialkyloxonium ion: R-O-R + H+ -> [R-O(H)-R]+, for example (C2H5)2O + HCl -> [(C2H5)2OH]+ Cl-. Formation of oxonium salts in concentrated acids explains why ethers dissolve in concentrated H2SO4 and how ether linkages are cleaved by hot concentrated HX.

Descriptive Questions

Question: 1(a). (a) What are alcohols? How alcohols are they classified?

Answer: Alcohols are organic compounds containing one or more hydroxyl (-OH) groups attached to an aliphatic carbon atom, general formula R-OH or CnH2n+1OH. They are classified in two ways: by the number of -OH groups, as monohydric (one -OH, e.g. ethanol), dihydric (two -OH, e.g. ethylene glycol) and trihydric/polyhydric (three or more, e.g. glycerol); and by the carbon bearing the -OH group, as primary (1 degree), secondary (2

degree) and tertiary (3 degree) alcohols.

Question: 1(b). (b) Starting from Grignard reagent how is primary, secondary and tertiary alcohol prepared?

Answer: A primary alcohol is obtained by reacting a Grignard reagent with formaldehyde and hydrolysing the product: RMgX + HCHO -> R-CH2-OMgX -> R-CH2OH (with H2O), the alcohol having one carbon more than R. A secondary alcohol is obtained from any other aldehyde: RMgX + R-CHO -> R2CH-OH after hydrolysis. A tertiary alcohol is obtained from a ketone: RMgX + R-CO-R -> R3C-OH after hydrolysis, the carbonyl carbon ending up attached to three carbon groups and one -OH group.

Question: 2(i). Write the equations for the following possible conversions. (i) Ethyl alcohol to diethyl ether

Answer: Ethanol is heated with concentrated H2SO4 at about 140 C, when two molecules lose one molecule of water between them (intermolecular dehydration). The equation is: 2C2H5OH -> C2H5-O-C2H5 + H2O (conc. H2SO4, 140 C). Excess ethanol and the lower temperature favour ether formation; at 170 C ethene is formed instead.

Question: 2(ii). Write the equations for the following possible conversions. (ii) Phenol to benzoquinone

Answer: Phenol is oxidised by a strong oxidising agent such as acidified sodium dichromate (or chromic acid) to p-benzoquinone, a yellow solid. The equation is: C6H5OH + [O] -> O=C6H4=O (p-benzoquinone) + H2O, in the presence of Na2Cr2O7/H2SO4. The -OH group is oxidised and a second carbonyl appears at the para position of the ring.

Question: 2(iii). Write the equations for the following possible conversions. (iii) Ethyl bromide to ethanol

Answer: Ethyl bromide is hydrolysed by boiling with aqueous KOH (or with moist Ag2O), the bromine being replaced by -OH. The equation is: C2H5Br + KOH (aq.) -> C2H5OH + KBr. Aqueous alkali must be used, because alcoholic KOH would give ethene by elimination.

Question: 2(iv). Write the equations for the following possible conversions. (iv) 2 degree-alcohol to carboxylic acid

Answer: A secondary alcohol is first oxidised to a ketone with acidified K2Cr2O7, and on vigorous oxidation with concentrated HNO3 or hot strong dichromate the ketone undergoes C-C bond cleavage to give carboxylic acids with fewer carbon atoms. Example: CH3-CH(OH)-CH3 -> ([O], K2Cr2O7/H2SO4) CH3-CO-CH3 (propanone) -> (vigorous [O], cleavage) CH3COOH + CO2/HCOOH products, ethanoic acid being the main acid obtained from propan-2-ol.

Question: 3(i). Differentiate between alcohol and phenol on the basis of (i) Solubility in water

Answer: Lower alcohols such as ethanol are completely miscible with water because the small alkyl group allows extensive hydrogen bonding with water. Phenol is only sparingly soluble in cold water (about 8 g per 100 mL) because its large non-polar benzene ring opposes dissolution, although it becomes fully miscible above about 66 C. Thus alcohols are generally more soluble in water than phenol.

Question: 3(ii). Differentiate between alcohol and phenol on the basis of (ii) Boiling Point

Answer: Phenol boils at about 182 C, much higher than ethanol (78 C), partly because of its larger molecular mass and strong hydrogen bonding combined with ring interactions. Alcohols also have high boiling points for their mass due to hydrogen bonding, increasing with chain length (methanol 65 C, ethanol 78 C, propan-1-ol 97 C). In comparable terms phenol, being aromatic and heavier, has a distinctly higher boiling point than the common lower alcohols.

Question: 3(iii). Differentiate between alcohol and phenol on the basis of (iii) Acidic character

Answer: Phenol is measurably acidic (pKa about 10): it turns blue litmus red, reacts with NaOH to form sodium phenoxide and water, and its phenoxide ion is stabilised by resonance delocalisation into the ring. Alcohols are essentially neutral (pKa about 16-18), do not affect litmus and do not react with NaOH, because the alkoxide ion gets no such resonance stabilisation and the alkyl group pushes electrons onto oxygen. Phenol is therefore a much stronger acid than an alcohol, though weaker than carboxylic acids and carbonic acid.

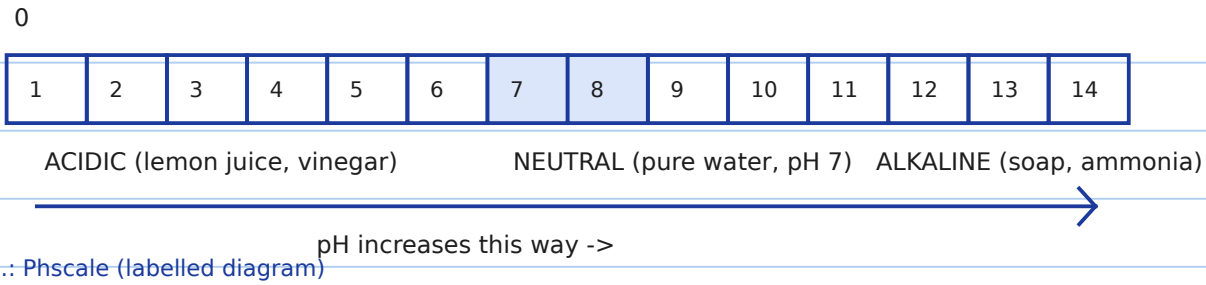


Fig.: Phscale (labelled diagram)

Question: 4(i). Write the equation and name the final product when phenol reacts with the following (i) Hot and concentrated nitric acid.

Answer: With hot concentrated nitric acid, phenol is both nitrated and oxidised, the main isolated product being 2,4,6-trinitrophenol, commonly called picric acid, a yellow crystalline explosive. The equation is: C6H5OH + 3HNO3 (conc., heat) -> C6H2(NO2)3-OH (picric acid) + 3H2O. In practice picric acid is usually made via

sulphonation first to reduce oxidation losses.

Question: 4(ii). Write the equation and name the final product when phenol reacts with the following (ii) Concentrated sulphuric acid at 100 C

Answer: With concentrated H2SO4 at about 100 C phenol is sulphonated mainly at the para position, giving p-phenol sulphonic acid as the major product (the ortho product predominates at room temperature). The equation is: C6H5OH + H2SO4 (conc., 100 C) -> p-HO-C6H4-SO3H + H2O. The final product is 4-hydroxybenzene sulphonic acid (p-phenolsulphonic acid).

Question: 4(iii). Write the equation and name the final product when phenol reacts with the following (iii) Bromine water

Answer: Phenol instantly decolourises bromine water and gives a white precipitate of 2,4,6-tribromophenol, because the -OH group strongly activates the ring at both ortho positions and the para position. The equation is: C6H5OH + 3Br2 -> C6H2Br3-OH (2,4,6-tribromophenol) + 3HBr. This reaction is used as a confirmatory test for phenol.

Question: 4(iv). Write the equation and name the final product when phenol reacts with the following (iv) Sodium metal

Answer: Phenol reacts with sodium metal to form sodium phenoxide with evolution of hydrogen gas, behaving like a weak acid. The equation is: 2C6H5OH + 2Na -> 2C6H5ONa (sodium phenoxide) + H2. The reaction is less vigorous than that of sodium with water or with carboxylic acids, in keeping with the weak acidity of phenol.

Question: 5. Enlist the commercial applications of Alcohol, Phenol and

Ether

Answer: Alcohols: ethanol is used as a fuel and fuel additive, as a solvent for paints, varnishes, perfumes and medicines, in alcoholic drinks and as an antiseptic; methanol is used to make formaldehyde, plastics and as an industrial solvent, and ethylene glycol is used as antifreeze. Phenol is used to make Bakelite and other phenol-formaldehyde resins, nylon intermediates, dyes, antiseptics such as TCP and Dettol-type products, aspirin and picric acid. Ethers: diethyl ether is used as a solvent for oils, fats and Grignard reactions and was used as a general anaesthetic, while other ethers serve as solvents and starting materials in industry.

Chapter 8: Carbonyl Compounds I - Aldehydes and Ketones

Multiple Choice Questions

1. (i) Ketones when treated with LiAlH4, they reduce to:

(a) Primary alcohol

✓ (b) Secondary alcohol

(c) Tertiary alcohol

(d) Dihyderic alcohol

Answer: (b) Secondary alcohol - LiAlH4 is a strong reducing agent that adds hydride to the carbonyl carbon. A ketone R-CO-R on reduction gives R-CHOH-R, in which the -OH is attached to a carbon bonded to two other carbons, so the product is a secondary alcohol.

2. (ii) The reagent used to oxidizes ketones into carboxylic acids is:

(a) Ammoniacal silver nitrate

✓ (b) Potassium dichromatic

(c) Fehling solution

(d) Benedict solution

Answer: (b) Potassium dichromatic - Ketones resist mild oxidising agents like Tollens reagent, Fehling solution and Benedict solution. Only a strong oxidising agent such as acidified potassium dichromate K2Cr2O7/H2SO4 can break the C-C bond next to the carbonyl group and oxidise a ketone into carboxylic acids with fewer carbon atoms.

3. (iii) The carbonyl carbon of aldehydes and ketones is:

(a) Sp hyrbidized

✓ (b) Sp2 hyrbidized

(c) Sp3 hyrbidized

(d) dsp3 hyrbidized

Answer: (b) Sp2 hyrbidized - The carbonyl carbon forms three sigma bonds (to oxygen and to two other atoms) in one plane and one pi bond to oxygen. Three sigma bonds in a trigonal planar arrangement means the carbon is sp2 hybridised, with bond angles of about 120 degrees.

4. (iv) Acetophenone is the member of ketone family, it contains:

(a) Two alkyl groups

(b) Two aryl group

✓ (c) One alkyl and one aryl group

(d) One aryl and one hydrogen atom

Answer: (c) One alkyl and one aryl group - Acetophenone is C6H5-CO-CH3, in which the carbonyl carbon is attached to a phenyl (aryl) group on one side and a methyl (alkyl) group on the other side. A hydrogen on the carbonyl carbon would make it an aldehyde, not a ketone.

5. (v) The most reactive molecule towards nucleophilic addition in the following is:

✓ (a) Formaldehyde

(b) Acetaldehyde

(c) Diethyl ketone

(d) Acetophenone

Answer: (a) Formaldehyde - Formaldehyde HCHO has two small hydrogen atoms on the carbonyl carbon, so there is almost no steric hindrance and no electron-donating alkyl group to reduce the positive charge on carbon.

A nucleophile can therefore attack its carbonyl carbon most easily, making it the most reactive of the four.

6. (vi) Clemmensen reduction is the conversion of aldehydes and ketones into:

✓ (a) Alkanes

(b) Alkenes

(c) Alkyl halides

(d) Alcohols

Answer: (a) Alkanes - In Clemmensen reduction the carbonyl group C=O of an aldehyde or ketone is completely reduced to a CH2 group by zinc amalgam Zn-Hg and concentrated HCl. For example CH3CHO is converted into CH3CH3, so the overall product is an alkane.

7. (vii) Hydration of propyne in the presence of H2SO4 and HgSO4 gives:

(a) Formaldehyde

(b) Methyl ethyl ketone

✓ (c) Acetone

(d) Acetaldehyde

Answer: (c) Acetone - Water adds to propyne CH3-C=CH (triple bond) according to Markovnikov rule in the presence of H2SO4/HgSO4, giving an unstable enol CH3-C(OH)=CH2. This enol tautomerises to the more stable ketone, acetone CH3-CO-CH3.

8. (viii) Which of the following carbonyl compound is most soluble in water?

✓ (a) Formaldehyde

(b) Acetaldehyde

(c) Benzaldehyde

(d) Acetophenone

Answer: (a) Formaldehyde - Formaldehyde is the smallest carbonyl compound and its oxygen forms strong hydrogen bonds with water, while it has no bulky hydrocarbon part to oppose dissolving. As the size of the alkyl or aryl group increases, as in benzaldehyde and acetophenone, solubility in water decreases, so formaldehyde is the most soluble.

9. (ix) Which of the following gives silver test with Tollens reagent?

✓ (a) HCHO

(b) CH3-O-CH3

(c) C2H5OH

(d) CH3COOH

Answer: (a) HCHO - Tollens reagent (ammoniacal silver nitrate) is a mild oxidising agent that is reduced by aldehydes only, depositing a silver mirror. Formaldehyde HCHO is an aldehyde, whereas dimethyl ether, ethanol and acetic acid do not reduce Tollens reagent.

10. (x) On reduction of a carbonyl compound by Zn-Hg and Conc. HCl, it is converted to an alkane. This reaction is known as:

(a) Dow reduction

(b) Cope reduction

✓ (c) Clemmensen reduction

(d) Wolf-Kishner reduction

Answer: (c) Clemmensen reduction - Reduction of aldehydes and ketones to alkanes with zinc amalgam and concentrated hydrochloric acid under strongly acidic conditions is called Clemmensen reduction. Wolf-Kishner reduction also gives alkanes but uses hydrazine NH2NH2 and strong base

KOH instead.

Short Questions

Question: 11. 1. Give reasons for the following: i. The boiling point of aldehydes and ketones is lower than alcohol. ii. Formaldehyde is highly soluble in water as compared to other aldehydes. iii. Oxidation of aldehydes is faster than ketones.

Answer: i. Alcohols have -OH groups and form strong intermolecular hydrogen bonds, while aldehydes and ketones have no hydrogen attached to oxygen, so they are held only by weaker dipole-dipole forces and need less heat to boil. ii. Formaldehyde is the smallest aldehyde; its carbonyl oxygen hydrogen-bonds strongly with water and there is no large non-polar hydrocarbon part, whereas in higher aldehydes the growing alkyl chain makes the molecule less polar and less soluble. iii. An aldehyde has a hydrogen on the carbonyl carbon which is easily removed during oxidation to a carboxylic acid, but a ketone has no such hydrogen and oxidation requires breaking a strong C-C bond, so aldehydes oxidise much faster and even with mild oxidising agents.

Question: 12. 2. Write the equation for the reaction of acetaldehyde with the following: i. Chromic acid (H2CrO4) ii. Lithium Aluminum hydride (LiAlH4) iii. Zinc mercury amalgam

Answer: i. With chromic acid acetaldehyde is oxidised to acetic acid: CH3CHO + [O] --H2CrO4--> CH3COOH. ii. With LiAlH4 it is reduced to ethanol, a primary alcohol: CH3CHO + 2[H] --LiAlH4--> CH3CH2OH. iii. With Zn-Hg and conc. HCl (Clemmensen reduction) the C=O is reduced to CH2 and ethane is formed: CH3CHO + 4[H] --Zn-Hg/conc. HCl--> CH3CH3 + H2O.

Question: 13. 3. How is formaldehyde prepared by ozonolysis?

Answer: Formaldehyde is obtained by the ozonolysis of ethene. Ethene reacts with ozone to form an unstable ozonide, which on hydrolysis with zinc and water breaks at the double bond: CH2=CH2 + O3 --> ozonide --Zn/H2O--> 2HCHO + H2O. Each CH2 end of the double bond becomes one molecule of formaldehyde, and zinc prevents its further oxidation to formic acid.

Question: 14. 4. Why is formaldehyde more reactive towards Nucleophilic addition reactions compared with ketones.

Answer: In formaldehyde the carbonyl carbon carries two hydrogen atoms, which are very small and do not donate electrons strongly, so the carbon carries a larger partial positive charge and a nucleophile can approach it without hindrance. In ketones two bulky alkyl groups donate electrons towards the carbonyl carbon, reducing its positive charge, and they also block the approach of the nucleophile sterically. Hence formaldehyde is far more reactive than ketones towards nucleophilic addition.

Question: 15. 5. How does the oxidation of ketones differ from the oxidation of aldehydes?

Answer: Aldehydes are oxidised easily, even by mild oxidising agents such as Tollens reagent, Fehling solution and Benedict solution, and give a carboxylic acid with the same number of carbon atoms, e.g. CH3CHO --> CH3COOH. Ketones do not react with these mild reagents and are oxidised only by strong oxidising agents like acidified K2Cr2O7 with C-C bond cleavage. Their oxidation products are carboxylic acids with fewer carbon atoms than the original ketone.

Question: 16. 6. Why are ethers considered as good solvents in organic reactions?

Answer: Ethers are good solvents because they dissolve a wide range of organic compounds, both polar and non-polar, yet they are chemically quite unreactive towards acids, bases, oxidising and reducing agents under normal conditions. They do not react with Grignard reagents and organometallic compounds, which is why reactions like the preparation of RMgX are carried out in dry ether. Their low boiling points also make them easy to remove after the reaction.

Descriptive Questions

Question: 17. 1. What are aldehydes and ketones? Describe the structure and type of hybridization in them.

Answer: Aldehydes are organic compounds containing the -CHO group, in which the carbonyl carbon is attached to at least one hydrogen, general formula R-CHO, while ketones contain the C=O group between two carbon groups, general formula R-CO-R. In both, the carbonyl carbon is sp2 hybridised: it forms three sigma bonds in a trigonal planar arrangement with angles of about 120 degrees, and the fourth valence forms a pi bond with oxygen using the unhybridised p-orbital. The C=O bond is polar because oxygen is more electronegative, so carbon carries a partial positive charge and oxygen a partial negative charge, which explains their nucleophilic addition reactions.

Question: 18. 2. Explain the acid-catalyzed and base-catalyzed nucleophilic addition reactions in aldehydes and ketones.

Answer: In base-catalysed addition, used with strong nucleophiles and weak nucleophilic reagents in basic medium, the nucleophile Nu- attacks

the carbonyl carbon first, forming an alkoxide intermediate, which then takes a proton from water to give the addition product, e.g. HCHO + HCN --base--> HCH(OH)CN. In acid-catalysed addition, used with weak nucleophiles like water, alcohols and ammonia, the acid first protonates the carbonyl oxygen, which increases the positive charge on carbon and makes it more attractive for the weak nucleophile. The nucleophile then attacks the carbon, and loss of a proton gives the final product such as a hemiacetal. Thus acid activates the carbonyl compound, while base activates or supplies a strong nucleophile.

Question: 19. 3. Describe how aldehydes are distinguished from ketones by the following laboratory test. (i) Tollens reagents (ii) Fehling solution

Answer: (i) Tollens reagent is ammoniacal silver nitrate [Ag(NH3)2]OH. On warming with an aldehyde it is reduced to metallic silver, which deposits as a bright silver mirror on the test tube: RCHO + 2[Ag(NH3)2]OH --> RCOOH + 2Ag + 4NH3 + H2O; ketones give no mirror. (ii) Fehling solution is an alkaline solution of copper sulphate complexed with tartrate. On heating with an aldehyde the blue Cu2+ is reduced to a red precipitate of cuprous oxide Cu2O: RCHO + 2Cu(OH)2 + NaOH --> RCOONa + Cu2O + 3H2O; ketones give no red precipitate. Both tests therefore identify an aldehyde and distinguish it from a ketone.

Question: 20. 4. Write the equation and give the name of major product in the following chemical process. i. Oxidation of acetone with acidified K2Cr2O7. ii. Reduction of acetaldehyde with NaBH4. iii. Hydration of ethyne in the presence of H2SO4/HgSO4 iv. Acylation of benzene in the presence of AlCl3.

Answer: i. Acetone is cleaved by strong oxidation to acetic acid and formic

acid (formic acid further oxidises to CO2 and H2O): CH3COCH3 + 4[O] --K2Cr2O7/H2SO4--> CH3COOH + HCOOH. Major carboxylic acid product is acetic acid. ii. CH3CHO + 2[H] --NaBH4--> CH3CH2OH; major product is ethanol. iii. HC=CH (triple bond) + H2O --H2SO4/HgSO4--> CH3CHO; major product is acetaldehyde (ethanal), formed through the unstable vinyl alcohol enol. iv. C6H6 + CH3COCl --anhydrous AlCl3--> C6H5COCH3 + HCl; major product is acetophenone, formed by Friedel-Crafts acylation.

Question: 21. 5. Give four differences between aldehydes and ketones.

Answer: First, an aldehyde has the carbonyl group at the end of the chain with at least one hydrogen attached (R-CHO), while a ketone has the carbonyl group in the middle, attached to two carbon groups (R-CO-R). Second, aldehydes are easily oxidised to carboxylic acids by mild oxidising agents, while ketones resist mild oxidising agents and need strong ones. Third, aldehydes give Tollens silver mirror test and Fehling red precipitate test, while ketones give neither test. Fourth, aldehydes are more reactive towards nucleophilic addition than ketones because they have less steric hindrance and a more positively charged carbonyl carbon.

Question: 22. 6. Write the equation for the nucleophilic addition reaction if formaldehyde treated with i. Hydrogen cyanide ii. Primary alcohol iii. Methyl magnesium bromide iv. Ammonia

Answer: i. With HCN in the presence of a base catalyst, formaldehyde gives its cyanohydrin, hydroxyacetonitrile (glycolonitrile): HCHO + HCN --> CH2(OH)CN. ii. With a primary alcohol ROH in acid catalysis it first gives a hemiacetal and then an acetal: HCHO + 2ROH --> CH2(OR)2 + H2O. iii. With methyl magnesium bromide it gives, after hydrolysis, ethanol, a primary alcohol with one extra carbon: HCHO + CH3MgBr -->

CH3CH2OMgBr --H2O/H+--> CH3CH2OH. iv. With ammonia, formaldehyde does not give a simple imine; six molecules condense with four of ammonia to form hexamethylenetetramine (urotropine): 6HCHO + 4NH3 --> (CH2)6N4 + 6H2O.

Question: 23. 7. Explain the factors that influence the reactivity of carbonyl compounds towards nucleophilic addition reaction.

Answer: Reactivity depends mainly on the charge on the carbonyl carbon and on steric hindrance around it. Electron-donating alkyl groups reduce the partial positive charge on carbon and make attack harder, so reactivity decreases in the order formaldehyde, then other aldehydes, then ketones. Bulky groups near the carbonyl carbon block the approach of the nucleophile, so larger alkyl or aryl groups lower the reactivity, as in acetophenone. Strong nucleophiles and base catalysis favour addition with weak carbonyl compounds, while acid catalysis, by protonating the oxygen, increases the positive charge on carbon and helps weak nucleophiles to attack.

Chapter 9: Carbonyl Compounds II - Carboxylic Acid and Functional Derivatives

Multiple Choice Questions

1. (i) The most common compound found in pineapple is:

(a) Acetic acid

(b) Ethanol

(c) Acetone

✓ (d) Ethyl butanoate

Answer: (d) Ethyl butanoate - The characteristic fruity smell and flavour of pineapple is due to the ester ethyl butanoate present in it. Many fruits get their pleasant odours from such low molar mass esters, not from free acids like acetic acid.

2. (ii) Two molecules of acetic acid on condensation gives:

(a) Ethyl acetate

(b) Aceticamide

(c) Acylhalide

✓ (d) Acetic anhydride

Answer: (d) Acetic anhydride - On strong heating, or with a dehydrating agent like P2O5, two molecules of acetic acid lose one molecule of water and join through an oxygen atom: 2CH3COOH --> (CH3CO)2O + H2O. The product (CH3CO)2O is acetic anhydride, an acid anhydride.

3. (iii) Carboxylic acid is stronger acid than:

(a) HCl

(b) HNO3

✓ (c) C2H5OH

(d) H2SO4

Answer: (c) C2H5OH - Carboxylic acids are weak acids compared with mineral acids like HCl, HNO3 and H2SO4, because they ionise only partially. They are, however, much stronger acids than ethanol C2H5OH, because the carboxylate ion RCOO- formed after losing H+ is stabilised by resonance, while the ethoxide ion is not.

4. (iv) The reagent that cannot produce an acyl halide in reaction with a carboxylic acid is:

(a) PCl3

(b) PCl5

✓ (c) HCl

(d) SOCl2

Answer: (c) HCl - Acyl halides are prepared by replacing the -OH of a carboxylic acid with chlorine using PCl3, PCl5 or thionyl chloride SOCl2, for example RCOOH + SOCl2 --> RCOCl + SO2 + HCl. Hydrochloric acid HCl alone cannot do this replacement, because it is not a chlorinating agent for the hydroxyl group.

5. (v) Benzoic acid is the product of oxidation of:

(a) Benzene

✓ (b) Ethyl Benzene

(c) Aniline

(d) Phenol

Answer: (b) Ethyl Benzene - Alkyl benzenes with a benzylic carbon are oxidised by strong oxidising agents like alkaline KMnO4 or acidified

K2Cr2O7 to benzoic acid, whatever the length of the side chain: C6H5CH2CH3 --[O]--> C6H5COOH. Benzene itself, aniline and phenol do not give benzoic acid on such oxidation.

6. (vi) Formation of acyl halide and amide by carboxylic acid involved

(a) Replacement of hydrogen

(b) Replacement of carbonyl group

✓ (c) Replacement of hydroxyl group

(d) Replacement of oxygen

Answer: (c) Replacement of hydroxyl group - All functional derivatives of carboxylic acids are formed by replacing the -OH group of -COOH by another group. Replacement by Cl gives an acyl halide RCOCl, by NH2 gives an amide RCONH2, by OR gives an ester and by OCOR gives an anhydride.

7. (vii) Formic acid is naturally found in:

(a) Venom of ants

✓ (b) Bees string

(c) Vinegar

(d) Butter

Answer: (b) Bees string - According to the book, formic acid (methanoic acid, HCOOH) occurs in the sting of bees, from where it enters the skin and causes pain and irritation. It is also associated with ant venom, vinegar contains acetic acid, and butter contains butyric acid.

8. (viii) Among the following compounds, the one with the highest boiling point is:

(a) Ethanol

(b) Acetaldehyde

✓ (c) Acetic acid

(d) Ethyl chloride

Answer: (c) Acetic acid - Acetic acid molecules pair up into cyclic dimers held by two strong hydrogen bonds, so much more heat is needed to separate them. Its boiling point (118 degrees C) is therefore higher than that of ethanol (78 degrees C), acetaldehyde (20 degrees C) and ethyl chloride (12 degrees C).

9. (ix) The formula of caproic acid is:

(a) C_4H_9COOH

✓ (b) $C_5H_{11}COOH$

(c) $C_6H_{13}COOH$

(d) $C_7H_{15}COOH$

Answer: (b) $C_5H_{11}COOH$ - Caproic acid is the six-carbon saturated fatty acid, hexanoic acid, found in goat fat. Counting the carboxyl carbon, its formula is C_5H_{11}-COOH, that is $C_6H_{12}O_2$ in total.

10. (x) The reaction of acetic acid with ethanol in the presence of conc. sulphuric acid gives:

✓ (a) Ethyl acetate

(b) Acetamide

(c) Ethane

(d) Acetic anhydride

Answer: (a) Ethyl acetate - This is esterification, in which the acid and the alcohol combine with loss of water in the presence of conc. H_2SO_4: $CH_3COOH + C_2H_5OH \xrightarrow{\text{conc. } H_2SO_4} CH_3COOC_2H_5 + H_2O$. The ester formed, ethyl acetate (ethyl ethanoate), has a sweet fruity smell.

Short Questions

Question: 11. 1. Explain why? i) The boiling points of carboxylic acids are high than alcohol? ii) The structure of carboxylic acid is trigonal planar?

Answer: i) Carboxylic acids are more polar than alcohols and form very strong hydrogen bonds between two molecules, giving stable cyclic dimers, so a large amount of heat energy is needed to break these bonds and boil the acid. Alcohols form weaker, single hydrogen-bonded associations, hence acids of comparable mass boil higher than alcohols. ii) The carboxyl carbon is sp2 hybridised and forms three sigma bonds - to R, to =O oxygen and to -OH oxygen - which arrange themselves in one plane at about 120 degrees to each other. This planar arrangement of three groups around the carbon makes the -COOH structure trigonal planar.

Question: 12. 2. What happens when: i) Formaldehyde reacts with a mixture of Potassium dichromate and sulphuric acid. ii) Carboxylic acid reacts with thionyl chloride iii) Carboxylic acid reacts with ammonia iv) Ethyl magnesium bromide reacts with carbon dioxide

Answer: i) Formaldehyde is fully oxidised, first to formic acid and then to carbon dioxide and water: HCHO --K2Cr2O7/H2SO4--> HCOOH --> CO2 + H2O, and the orange dichromate turns green. ii) The -OH of the acid is replaced by Cl and an acyl chloride is formed: RCOOH + SOCl2 --> RCOCl + SO2 + HCl. iii) The acid first forms an ammonium salt, which on heating loses water to give an amide: RCOOH + NH3 --> RCOONH4 --heat--> RCONH2 + H2O. iv) The Grignard reagent adds to CO2 (carbonation) and on acid hydrolysis gives propanoic acid, one carbon more than ethyl group: C2H5MgBr + CO2 --> C2H5COOMgBr --H3O+--> C2H5COOH.

Question: 13. 3. Write down the commercial applications of carboxylic

acids.

Answer: Carboxylic acids are used as preservatives, flavouring agents and acidity regulators in the food industry, for example acetic acid as vinegar and benzoic acid as a preservative. They are starting materials for making soaps (salts of long-chain fatty acids), detergents, perfumes and esters used as solvents and flavours. They are also used in making medicines such as aspirin, in the manufacture of polymers like nylon and polyester, and in the rubber, textile dyeing and leather industries.

Question: 14. 4. Write the natural sources of following carboxylic acids. (a) Formic acid (b) Acetic acid (c) Valeric acid (d) Caproic acid

Answer: (a) Formic acid occurs in the sting of bees and in ant venom. (b) Acetic acid is present in vinegar, formed by the fermentation of ethanol in air. (c) Valeric acid (pentanoic acid) occurs in the valerian plant root, from which it takes its name. (d) Caproic acid (hexanoic acid) occurs in goat milk and goat fat (capra means goat) and in small amounts in other animal fats.

Descriptive Questions

Question: 15. 1. How is carboxylic acid prepared by: i) Carbonation of Grignard reagent ii) Hydrolysis of alkyl nitriles iii) Oxidation of primary alcohols

Answer: i) A Grignard reagent RMgX is treated with dry carbon dioxide (dry ice); the reagent adds to CO2 and the adduct on hydrolysis with dilute acid gives a carboxylic acid with one carbon more than R: RMgX + CO2 --> RCOOMgX --H3O+--> RCOOH. ii) An alkyl nitrile RCN is heated with dilute acid or alkali; it hydrolyses first to an amide and then to the acid: RCN + 2H2O --H+/heat--> RCOOH + NH3. iii) A primary alcohol is oxidised with a

strong oxidising agent like acidified K2Cr2O7 or KMnO4; it passes through the aldehyde stage to the acid with the same number of carbons: RCH2OH --[O]--> RCHO --[O]--> RCOOH, for example ethanol gives acetic acid.

Question: 16. 2. Explain the structure of carboxylic acid.

Answer: A carboxylic acid contains the carboxyl group -COOH, which is a combination of a carbonyl group C=O and a hydroxyl group -OH on the same carbon. This carbon is sp2 hybridised, so the group is trigonal planar with bond angles near 120 degrees, and the lone pairs on oxygen conjugate with the carbonyl pi bond, giving resonance in the group. Because of the polar C=O and the -OH, two molecules join by two hydrogen bonds into a cyclic dimer, which explains the high boiling points and association of carboxylic acids.

Question: 17. 3. Discuss the acidic nature of carboxylic acid. How is it stronger than other organic compounds and weaker than mineral acids?

Answer: Carboxylic acids turn blue litmus red, react with metals, carbonates and alkalis, and lose the hydroxyl proton in water: RCOOH + H2O --> RCOO- + H3O+. They are stronger than alcohols and phenols because the carboxylate ion RCOO- is stabilised by resonance, the negative charge being shared equally by both oxygens, so the proton is lost easily. They are weaker than mineral acids like HCl and H2SO4 because their ionisation is only partial (acetic acid is only a few percent ionised) - the equilibrium lies to the left, whereas mineral acids ionise almost completely in water.

Question: 18. 4. Convert the followings: i) Carboxylic acid into acid anhydride ii) Ester into carboxylic acid iii) Toluene into benzoic acid

Answer: i) Heat two molecules of the acid with a strong dehydrating agent such as P2O5; one water molecule is removed: 2CH3COOH --P2O5/heat--> (CH3CO)2O + H2O, giving acetic anhydride. ii) Hydrolyse the ester by heating with dilute acid or alkali; with alkali the salt is formed, which on acidification gives the acid: CH3COOC2H5 + H2O --H+/heat--> CH3COOH + C2H5OH. iii) Oxidise the methyl side chain of toluene with strong alkaline KMnO4 followed by acidification, or acidified K2Cr2O7: C6H5CH3 --[O]--> C6H5COOH, giving benzoic acid.

Question: 19. 5. Explain the following physical properties of carboxylic acids: (a) Solubility (b) Boiling point

Answer: (a) The lower carboxylic acids up to four carbons are miscible or highly soluble in water because the -COOH group forms hydrogen bonds with water molecules. As the alkyl chain grows, the non-polar hydrocarbon part dominates, hydrogen bonding becomes less effective, and solubility falls rapidly, so higher fatty acids are practically insoluble in water though soluble in organic solvents. (b) Carboxylic acids have unusually high boiling points, higher than alcohols of similar mass, because their molecules associate into cyclic dimers through two strong hydrogen bonds. Extra energy is needed to break this dimer, so, for example, acetic acid boils at 118 degrees C while ethanol boils at 78 degrees C.

Question: 20. 6. Write the names of four derivatives of carboxylic acids and give the equation for their preparation from acetic acid.

Answer: The four derivatives are acetyl chloride (acyl halide), acetic anhydride (acid anhydride), ethyl acetate (ester) and acetamide (amide), all formed by replacing the -OH or H of acetic acid. Equations: CH3COOH + SOCl2 --> CH3COCl + SO2 + HCl (acetyl chloride); 2CH3COOH

--P2O5/heat--> (CH3CO)2O + H2O (acetic anhydride); CH3COOH + C2H5OH --conc. H2SO4--> CH3COOC2H5 + H2O (ethyl acetate); CH3COOH + NH3 --> CH3COONH4 --heat--> CH3CONH2 + H2O (acetamide).

Chapter 10: Biochemistry

Multiple Choice Questions

1. (i) Starch and Sucrose are examples of:

(a) Monosaccharides and Disaccharides

(b) Disaccharides and Oligosaccharides

✓ (c) Polysaccharides and Disaccharides

(d) Monosaccharides and Polysaccharides

Answer: (c) Polysaccharides and Disaccharides - Starch is made of very many glucose units joined together, so it is a polysaccharide used as energy storage in plants. Sucrose (table sugar) is made of only two units, glucose and fructose, so it is a disaccharide.

2. (ii) Amino acid units bonded in protein molecule through:

(a) Glycosidic linkage

(b) Ether linkage

✓ (c) Peptide linkage

(d) Hydrogen bridge

Answer: (c) Peptide linkage - In a protein, the -COOH group of one amino acid joins with the -NH2 group of the next amino acid with loss of a water molecule, forming a -CO-NH- bond. This amide bond between amino acids is called the peptide linkage, and long chains of such linkages make the protein.

3. (iii) Proteins are composed of:

✓ (a) Amino acids

(b) Carbohydrates

(c) Lipids

(d) Nucleic acids

Answer: (a) Amino acids - Proteins are natural condensation polymers whose building blocks are alpha amino acids. About 22 amino acids join in a specific sequence through peptide bonds to make different proteins, which on hydrolysis give back these amino acids.

4. (iv) A condensation polymer of amino acid is:

✓ (a) Protein

(b) Lipids

(c) Starch

(d) Glycogen

Answer: (a) Protein - Amino acids condense with one another, losing water molecules and forming peptide bonds, to give the long polypeptide chains called proteins. Starch and glycogen are condensation polymers of glucose, not of amino acids, and lipids are esters, not condensation polymers of amino acids.

5. (v) Saponification is the formation of soap by the reaction of fat and oil with:

✓ (a) An alkali

(b) An acid

(c) Sugar

(d) Glycerol

Answer: (a) An alkali - When a fat or oil (a triglyceride ester) is boiled with an alkali such as NaOH or KOH, it hydrolyses to glycerol and the sodium or potassium salts of fatty acids, which are soaps. This alkaline hydrolysis of fats is called saponification.

6. (vi) Which of the following mineral is considered to be essential for immune system:

(a) Iron

(b) Zinc ✓

(c) Magnesium

(d) Calcium

Answer: (b) Zinc - Zinc is essential for the normal development and working of immune cells, for enzyme activity and for wound healing. Its deficiency weakens immunity and makes a person more prone to infections, while iron mainly serves in haemoglobin and calcium in bones and teeth.

7. (vii) Rancidity is a chemical process involving:

(a) Oxidation and hydrolysis ✓

(b) Condensation and reduction

(c) Polymerization

(d) Decarboxylation

Answer: (a) Oxidation and hydrolysis - When fats and oils are exposed to air, light and moisture, their unsaturated parts are oxidised and the ester bonds are hydrolysed, producing short-chain acids and aldehydes with bad taste and smell. These two simultaneous reactions, oxidation and hydrolysis, together constitute rancidity.

8. (viii) Lipid which is a major component of cell membrane is:

(a) Triglyceride

(b) Phospholipid ✓

(c) Glycolipid

(d) Steroid

Answer: (b) Phospholipid - Cell membranes are built mainly of a double layer of phospholipids, whose hydrophilic phosphate heads face the water on both sides and whose hydrophobic fatty acid tails face inwards. Triglycerides are storage fats, while glycolipids and steroids are present in membranes only in smaller amounts.

9. (ix) Total numbers of alpha amino acids are:

(a) 19

✓ (b) 22

(c) 25

(d) 28

Answer: (b) 22 - According to the book, there are 22 alpha amino acids in total that take part in making proteins in living organisms. Of these, some can be made by the body itself while the essential ones must be supplied through diet.

10. (x) Sugar molecules are classified as:

(a) Fats

(b) Proteins

✓ (c) Carbohydrates

(d) Lipids

Answer: (c) Carbohydrates - Sugars such as glucose, fructose and sucrose are polyhydroxy aldehydes or ketones, or substances that give them on hydrolysis, which is the definition of carbohydrates. Fats and lipids are esters of fatty acids, and proteins are polymers of amino acids, so sugars belong only to the carbohydrate class.

Short Questions

Question: 11. 1. Mention the three main functions of lipids.

Answer: First, lipids act as a concentrated store of energy in the body and provide about twice as much energy per gram as carbohydrates. Second, they form the structural material of cell membranes, especially phospholipids, and protect delicate internal organs as a cushion. Third, they act as thermal insulators under the skin, prevent water loss, and serve as solvents and carriers for the fat-soluble vitamins A, D, E and K.

Question: 12. 2. Comparing with other nutrients, why lipids are better source of energy?

Answer: Lipids contain a much higher proportion of carbon and hydrogen and very little oxygen compared with carbohydrates and proteins, so they can be oxidised more completely. On complete oxidation 1 g of lipid gives about 9 kcal (about 38 kJ) of energy, while 1 g of carbohydrate or protein gives only about 4 kcal. Lipids are also stored in the body in nearly water-free form, so more energy is stored per unit weight, making them the better energy source.

Question: 13. 3. Carbohydrates are necessary component of our diet. Give two dietary importance of carbohydrates.

Answer: First, carbohydrates are the main and quickest source of energy for our tissues and organs; glucose is the immediate fuel, especially for the brain, and 1 g gives about 4 kcal. Second, they spare proteins from being used as fuel, store extra energy as glycogen in liver and muscles and as starch-rich foods, and provide dietary fibre (cellulose) that aids digestion and bowel movement.

Question: 14. 4. What is meant by saponification? Give the reaction.

Answer: Saponification is the hydrolysis of a fat or oil with an alkali to form glycerol and soap, which is the sodium or potassium salt of a long-chain fatty acid. In general: triglyceride (fat/oil) + 3NaOH --heat--> glycerol + 3RCOONa (soap). For example, glyceryl stearate boiled with NaOH gives glycerol and sodium stearate, a common soap.

Question: 15. 5. What is rancidity which chemical reaction involves in this process?

Answer: Rancidity is the spoiling of fats and oils on standing in air, in which they develop a bad taste and unpleasant smell and become unfit to eat. It involves two simultaneous reactions: oxidation of the unsaturated fatty acid parts by atmospheric oxygen, and hydrolysis of the ester bonds by moisture, helped by enzymes and micro-organisms. The products include volatile short-chain fatty acids and aldehydes that cause the bad odour.

Question: 16. 6. Write three essential functions of protein in the body.

Answer: First, proteins are the body-building material: they build and repair muscles, skin and other tissues and are essential for growth. Second, many proteins act as enzymes and hormones that control and speed up the chemical reactions and processes of the body, such as insulin and digestive enzymes. Third, proteins like antibodies defend the body against disease, haemoglobin carries oxygen, and in need proteins can also be oxidised to provide energy.

Question: 17. 7. Write down the sources from which we intake fructose and lactose.

Answer: Fructose, the fruit sugar, is taken in from sweet fruits such as

grapes, apples, mangoes and dates, from honey, and from table sugar sucrose which gives fructose on hydrolysis. Lactose, the milk sugar, is obtained only from milk and milk products such as curd, cheese and milk-based foods. On digestion lactose is hydrolysed to glucose and galactose.

Descriptive Questions

Question: 18. 1. What are Carbohydrates? Give their classification on the basis of structure.

Answer: Carbohydrates are polyhydroxy aldehydes and ketones, or substances that yield such compounds on hydrolysis; they are the main source of energy for the body, with sources like rice, potato, wheat and barley. On the basis of structure (number of sugar units) they are classified into four types. Monosaccharides contain a single sugar unit and cannot be hydrolysed further, e.g. glucose, fructose and galactose; disaccharides contain two units, e.g. sucrose, lactose and maltose; oligosaccharides contain 3 to 10 units, e.g. kestose and melezitose; and polysaccharides contain more than 10 units, e.g. starch, cellulose and glycogen.

Question: 19. 2. Explain the role of glucose, fructose, sucrose and lactose in the health of human being.

Answer: Glucose is the immediate source of energy for cells and the normal sugar of blood; its level is controlled by insulin, and its failure leads to diabetes while its shortage causes weakness. Fructose is the sweetest natural sugar, taken from fruits and honey, metabolised in the liver and providing energy, but excess intake burdens the liver. Sucrose, the common table sugar from sugarcane and beet, is hydrolysed to glucose and fructose and supplies quick energy, though excess causes obesity and

tooth decay. Lactose, the milk sugar, provides energy to infants and helps absorption of calcium, but people lacking the enzyme lactase cannot digest it properly (lactose intolerance).

Question: 20. 3. What are Proteins? Classify various types of proteins on the basis of their function.

Answer: Proteins are large biomolecules made by the condensation of alpha amino acids joined by peptide bonds, and they perform most of the work in living cells. On the basis of function they are classified as: structural proteins (collagen, keratin) that build tissues; enzymes (amylase, pepsin) that catalyse body reactions; hormones (insulin) that regulate processes; transport proteins (haemoglobin) that carry substances; protective proteins (antibodies) that fight infection; contractile proteins (actin, myosin) that cause muscle movement; and storage proteins (casein of milk) that store amino acids and nutrients.

Question: 21. 4. What are Lipids? Give their classification, properties and biological significance.

Answer: Lipids are water-insoluble, heterogeneous organic compounds which are chemically esters of fatty acids and alcohols, soluble in organic solvents like ether and chloroform. They are classified into simple lipids (fats, oils and waxes), compound lipids (phospholipids and glycolipids) and derived lipids (steroids, fatty acids, glycerol). Their properties include insolubility in water, greasy feel, lower density than water, hydrolysis by alkali (saponification) and becoming rancid on oxidation. Biologically they store concentrated energy, build cell membranes, insulate the body, protect organs, and carry fat-soluble vitamins and hormones.

Question: 22. 5. How can you explain primary, secondary and tertiary structure of proteins?

Answer: The primary structure is the specific sequence in which amino acids are joined by peptide bonds in the polypeptide chain; it decides all higher structures of the protein. The secondary structure is the regular folding of this chain into an alpha-helix or a beta-pleated sheet, held by hydrogen bonds between C=O and N-H groups of nearby peptide bonds. The tertiary structure is the overall three-dimensional folding of the whole chain upon itself, stabilised by hydrogen bonds, ionic bonds, disulphide bonds and hydrophobic interactions, and it is this final shape that gives the protein its biological activity.

Question: 23. 6. Describe physical properties of proteins.

Answer: Proteins are colourless, mostly tasteless, high molar mass macromolecular solids; some are crystalline while most are amorphous. Some proteins like albumin of egg are soluble in water forming colloidal solutions, while fibrous proteins like keratin are insoluble in water and common solvents. They are denatured (their structure is destroyed) by heat, strong acids, alkalis, alcohol and heavy metal salts, as seen when egg white sets on heating. Proteins also show optical activity, can be precipitated from solution, and on strong heating they char and give a characteristic burnt-hair smell.

Question: 24. 7. Why minerals are essential for our health? Give the biological significance of Calcium, Iron, Zinc, and phosphorus.

Answer: Minerals are inorganic elements needed in small amounts for building body structures and for regulating enzymes, fluids, nerves and energy processes, and their deficiency causes specific diseases. Calcium

builds bones and teeth, helps blood clotting and muscle contraction, and its deficiency causes weak bones. Iron is the central part of haemoglobin and carries oxygen in blood; its deficiency causes anaemia. Zinc is essential for the immune system, wound healing, growth and many enzymes. Phosphorus, with calcium, forms bones and teeth, and as part of ATP it stores and transfers energy, besides being a component of DNA, RNA and cell membranes.

Chapter 11: Industrial Chemistry

Multiple Choice Questions

1. (i) Super glue is chemically named as:

✓ (a) Cyano acrylate

(b) Polyvinyl acetate

(c) Epoxy resins

(d) Polyurethane

Answer: (a) Cyano acrylate - Super glue is based on cyanoacrylate (methyl or ethyl cyanoacrylate), which polymerises within seconds on contact with the moisture present on surfaces. It therefore forms an extremely strong bond almost instantly, unlike the slower-setting epoxy resins or polyvinyl acetate adhesives.

2. (ii) DDT is a chemical which commercially known as:

✓ (a) Insecticide

(b) Herbicide

(c) Pesticide

(d) Fungicide

Answer: (a) Insecticide - DDT (dichlorodiphenyltrichloroethane) was widely used commercially to kill insects, especially mosquitoes spreading malaria, so it is classed as an insecticide. Herbicides kill weeds and fungicides kill fungi, whereas DDT acts specifically against insects.

3. (iii) Nail polish remover is mainly consists of:

(a) Pigments

✓ (b) Acetone

(c) Diethyl ether

(d) Ethyl alcohol

Answer: (b) Acetone - Nail polish remover consists mainly of acetone, a powerful solvent that quickly dissolves the film-forming polymer and pigments of dried nail polish. Pigments are a component of the polish itself, not of the remover.

4. (iv) Antimalarial drug among the following is:

(a) Ibuprofin

✓ (b) Chloroquine

(c) Paracetamol

(d) Diphenyl hdramine

Answer: (b) Chloroquine - Chloroquine is the well-known drug used for the prevention and treatment of malaria caused by the Plasmodium parasite. Ibuprofen and paracetamol are pain relievers and fever reducers, and diphenhydramine is an antihistamine used for allergy.

5. (v) Asprin is a pain reliever, its chemical name is:

(a) Ascorbic acid

(b) Nicotinic acid

✓ (c) Acetyl salisylic acid

(d) Benzoic acid

Answer: (c) Acetyl salisylic acid - Aspirin is acetylsalicylic acid, prepared by acetylating salicylic acid with acetic anhydride. It acts as an analgesic (pain reliever), antipyretic (fever reducer) and anti-inflammatory drug, while ascorbic acid is vitamin C.

6. (vi) Nylon 6, 6 is a condensation polymer of hexamethylene diamine

and:

(a) Benzoic acid

(b) Adipic acid

(c) Pthalic acid

(d) Valeric acid

Answer: (b) Adipic acid - Nylon 6,6 is formed by the condensation polymerisation of hexamethylene diamine (6 carbons) with adipic acid (6 carbons), with loss of water molecules and formation of amide links. The two 6-carbon monomers give it the name nylon 6,6.

7. (vii) Which of the following is not a synthetic plastic:

(a) Nylon

(b) Teflon

(c) Cellulose

(d) Polyethene

Answer: (c) Cellulose - Cellulose is a natural polymer obtained from plants (wood and cotton) and is not made synthetically as a plastic. Nylon, Teflon and polyethene are all synthetic polymers manufactured in industry from petrochemical raw materials.

8. (viii) Drugs that lower the body temperature to normal are known as:

(a) Antibiotics

(b) Antipyretic

(c) Antiallergic

(d) Anti histamins

Answer: (b) Antipyretic - Drugs that bring a raised body temperature (fever) back to normal are called antipyretics, common examples being paracetamol and aspirin. Antibiotics kill or stop bacteria, while antiallergic

drugs and antihistamines treat allergy symptoms.

9. (ix) Which of the following chemical is used as an oxidizing agent in permanent hair dyes:

(a) Acetone

✓ (b) Hydrogen peroxide

(c) Polyvinyl acetate

(d) Resorcinol

Answer: (b) Hydrogen peroxide - In permanent hair dyes, hydrogen peroxide H_2O_2 acts as the oxidising agent (developer): it oxidises the dye precursors so that large coloured molecules form inside the hair, and it also lightens the natural pigment. Resorcinol is a dye coupler, not the oxidiser.

10. (x) An example of thermosetting plastic is:

(a) Polyethene

(b) PVC

(c) Nylon

✓ (d) Bakelite

Answer: (d) Bakelite - Bakelite, made from phenol and formaldehyde, forms a hard cross-linked network on heating and cannot be softened or remoulded again, which is the property of a thermosetting plastic. Polyethene, PVC and nylon are thermoplastics that soften on heating and can be remoulded.

Short Questions

Question: 11. 1. Give the scope of pharmaceutical industries in Pakistan.

Answer: The pharmaceutical industry of Pakistan manufactures a large

share of the medicines used in the country, including tablets, syrups, injections, antibiotics, painkillers and life-saving drugs, and it also exports medicines to many countries. It provides employment to a large number of chemists, pharmacists and workers and saves foreign exchange by reducing imports of finished drugs. Its scope is growing with increasing population and health needs, although it still depends largely on imported raw materials (active ingredients).

Question: 12. 2. What is antihistamine drug? Give the symptoms in which it is used.

Answer: An antihistamine is a drug that blocks the action of histamine, a chemical released in the body during allergic reactions. It is used for symptoms such as sneezing, running nose, itching, skin rashes and hives (urticaria), watery eyes and hay fever. Common examples are diphenhydramine and cetirizine; some older antihistamines also cause drowsiness and are used in cough and cold medicines.

Question: 13. 3. Write the names of two synthetic and two natural polymers.

Answer: Two synthetic polymers are polyethene (used for bags and bottles) and nylon (used for fibres and ropes), both manufactured in industry from petrochemical monomers. Two natural polymers are cellulose (the structural material of plant cell walls, in wood and cotton) and starch (the energy-storage polymer of plants). Natural rubber and proteins are other examples of natural polymers.

Question: 14. 4. Write the name of four main components of Nail polish?

Answer: The four main components of nail polish are: (1) a film-forming

polymer, usually nitrocellulose, which forms the hard glossy film on the nail; (2) a resin, which improves adhesion and gloss; (3) a plasticiser such as camphor or dibutyl phthalate, which keeps the film flexible and prevents cracking; and (4) solvents such as ethyl acetate or butyl acetate, plus pigments and colours that give the polish its shade.

Descriptive Questions

Question: 15. 1. How can you define Cosmetics? Describe four cosmetics which are commonly used.

Answer: Cosmetics are substances applied to the body, especially the face, skin, hair and nails, to clean, protect, beautify and improve appearance. Four commonly used cosmetics are: nail polish, a coloured lacquer of nitrocellulose, resin, plasticiser and solvent applied on nails; lipstick, a mixture of waxes, oils and pigments applied to the lips; perfumes, solutions of fragrant oils in alcohol that give a pleasant smell; and hair dyes, which colour the hair - permanent dyes use hydrogen peroxide as oxidiser. Other common cosmetics include face creams, lotions, powders and nail polish remover (acetone).

Question: 16. 2. What are pesticides? Explain various types of pesticides along with their specific use.

Answer: Pesticides are chemicals used to kill or control pests that damage crops, stored food and health. Their main types are: insecticides, which kill insects, e.g. DDT used against mosquitoes and crop insects; herbicides (weedicides), which kill unwanted weeds in fields, e.g. 2,4-D; fungicides, which kill fungi attacking crops, e.g. Bordeaux mixture; and rodenticides, which kill rats and mice damaging stored grain. Pesticides increase crop yield but must be used carefully because many are poisonous and persist

in the environment.

Question: 17. 3. Describe the preparation, properties and uses of Nylon and polyvinyl chloride.

Answer: Nylon 6,6 is prepared by condensation polymerisation of hexamethylene diamine with adipic acid with loss of water; it is a strong, elastic, abrasion-resistant polyamide fibre used for ropes, fabrics, carpets, brushes and machine parts. Polyvinyl chloride (PVC) is prepared by addition polymerisation of vinyl chloride CH2=CHCl: nCH2=CHCl --> -[CH2-CHCl]n-. PVC is hard, tough, chemically resistant, non-flammable and a good electrical insulator; it is used for water and drainage pipes, electric cable insulation, floor tiles, raincoats, bottles and plastic sheets.

Question: 18. 4. What are adhesives? Explain the significance of super glue and silicon resins.

Answer: Adhesives are substances used to join two surfaces together by bonding, for example glues, gums and resins used in woodwork, packaging and industry. Super glue (cyanoacrylate) is significant because it polymerises in seconds on contact with surface moisture and forms an extremely strong bond between almost any materials, even skin, so it is used for quick repairs and even in surgery. Silicone resins (the book writes silicon resins) are heat-resistant, water-repellent adhesives and sealants that remain flexible over a wide temperature range, used for sealing joints, in electronics and in high-temperature applications.

Chapter 12: Environmental Chemistry

Multiple Choice Questions

Question: 1. (i) The greenhouse gas is: (a) Oxygen (O2) (b) Nitrogen (N2) (c) Argon (Ar) (d) Carbon dioxide (CO2)

Answer: (d) Carbon dioxide (CO2) - Carbon dioxide absorbs the infrared (heat) radiation re-radiated from the Earth and prevents it from escaping into space, thus warming the atmosphere - the greenhouse effect. Oxygen, nitrogen and argon, the main components of air, do not absorb infrared radiation in this way.

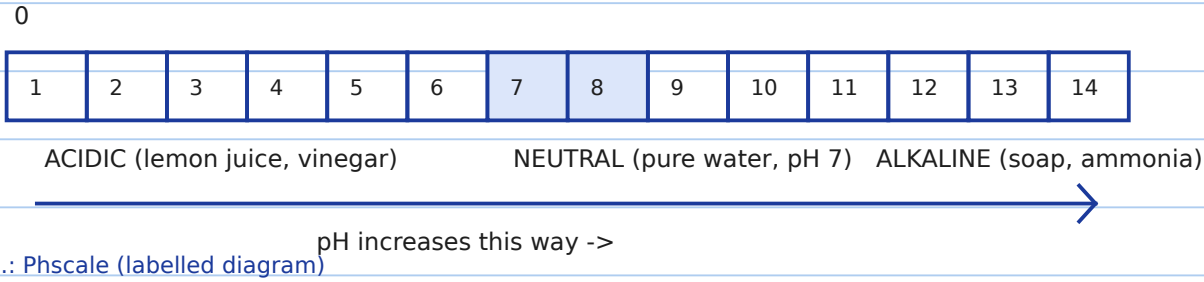


Fig.: Phscale (labelled diagram)

2. (ii) The pH of acid rain is:

(a) Between 7 to 8

(b) Between 6 to 7

✓ (c) Below 5

(d) Above 8

Answer: (c) Below 5 - Normal rain is slightly acidic (pH about 5.6) because of dissolved CO2, but acid rain contains sulphuric acid and nitric acid formed from SOx and NOx pollutants. This lowers its pH further, typically below 5 and often near 4, which harms soil, water life and buildings.

Question: 3. (iii) Ozone depletion in upper atmosphere is mainly caused by: (a) Sulphur dioxide (SO2) (b) Nitrogen oxides (NOx) (c) Carbon

monoxide (CO) (d) Chlorofluorocarbons (CFCs)

Answer: (d) Chlorofluorocarbons (CFCs) - CFCs used in refrigerators, air conditioners and sprays rise to the stratosphere, where UV radiation breaks them to release chlorine atoms. Each chlorine atom destroys very many ozone molecules by a chain reaction: Cl + O3 --> ClO + O2, so CFCs are the main cause of ozone depletion.

4. (iv) The region of sphere which extend from 11km to 50km from our Earths is known as:

(a) Troposphere

✓ (b) Stratosphere

(c) Mesosphere

(d) Thermosphere

Answer: (b) Stratosphere - The atmosphere is divided into layers: the troposphere extends from the surface to about 11 km, and above it the stratosphere extends from 11 km to 50 km. The stratosphere contains the ozone layer that absorbs harmful ultraviolet radiation; above it lie the mesosphere (50-85 km) and thermosphere (85-500 km).

5. (v) The increase in global average temperatures is primarily attributed to:

(a) Solar radiations

✓ (c) Green house gases emission

(d) Natural climate variations

Answer: (c) Green house gases emission - The rise in global average temperature (global warming) is mainly due to the increased emission of greenhouse gases such as CO2, CH4 and N2O from burning fossil fuels, industry and deforestation. These gases trap more of the outgoing infrared

radiation in the atmosphere, so the Earth warms up.

6. (vi) The primary goal of green chemistry is:

(a) Maximizing industrial production

(b) Minimizing waste generation

(c) Reducing energy consumption

✓ (d) Developing environmentally friendly chemicals

Answer: (d) Developing environmentally friendly chemicals - Green chemistry aims to design chemical products and processes that are safe for human health and the environment, preventing pollution at its source. Its goals include developing environmentally friendly chemicals, minimising waste and hazards, and using renewable raw materials and safer solvents.

7. (vii) Smog is word used for the combination of:

(a) Water vapours and frog

✓ (b) Smoke and fog

(c) Sediments and colloids

(d) Oxides of nitrogen and sulphur

Answer: (b) Smoke and fog - The word smog is made from smoke + fog. It is a dense, harmful mixture of fog with smoke and pollutant gases (SOx, NOx and unburnt hydrocarbons) that hangs over industrial cities, reduces visibility and causes breathing problems.

8. (viii) The catalyst used in automobile catalytic converter is a mixture of:

✓ (a) Pt and Pd

(b) Ni and Fe

(c) Cu and Cr

(d) Pb and Hg

Answer: (a) Pt and Pd - Catalytic converters use noble-metal catalysts, mainly platinum (Pt) and palladium (Pd), usually with rhodium, coated on a honeycomb support. They convert poisonous exhaust gases into harmless ones: 2CO + O2 --> 2CO2, unburnt hydrocarbons --> CO2 + H2O, and 2NOx --> N2 + xO2.

9. (ix) Ozone layer is present in which of the following region of atmosphere:

(a) Troposphere

✓ (b) Stratosphere

(c) Mesosphere

(d) Thermosphere

Answer: (b) Stratosphere - The ozone layer lies in the stratosphere, about 15 to 35 km above the Earth, where ozone O3 is formed from oxygen by ultraviolet radiation. This layer absorbs most of the harmful UV radiation of the Sun and protects life on Earth from it.

10. (x) The major source of water pollution is:

(a) Organic farming

(b) Global warming

✓ (c) Industrial activities

(d) Renewable energy sources

Answer: (c) Industrial activities - Industrial activities discharge untreated effluents containing toxic chemicals, heavy metals, dyes, acids and oils into rivers and groundwater, making industry the major source of water pollution. Domestic sewage and agricultural run-off of fertilisers and pesticides are the other major sources.

Short Questions

Question: 11. 1. Write the name of four segments of atmosphere and mention in which segment we live.

Answer: The four segments (layers) of the atmosphere are: the troposphere (surface to 11 km), the stratosphere (11 to 50 km), the mesosphere (50 to 85 km) and the thermosphere (85 to 500 km). We live in the lowest layer, the troposphere, which contains the air we breathe and where weather phenomena like rain, clouds and storms occur.

Question: 12. 2. Explain the causes of depletion of ozone layer.

Answer: The main cause is the release of ozone-depleting substances, especially chlorofluorocarbons (CFCs) from refrigerators, air conditioners, aerosol sprays and foam making, and also halons and nitrogen oxides. In the stratosphere, UV radiation breaks CFCs and releases chlorine atoms: CFCl3 --UV--> CFCl2 + Cl. The chlorine atom destroys ozone in a repeating chain: Cl + O3 --> ClO + O2 and ClO + O --> Cl + O2, so one chlorine atom can destroy about 100,000 ozone molecules, thinning the ozone layer.

Question: 13. 3. What is industrial smog and how is it formed?

Answer: Industrial smog is the grey, harmful smog found over industrial cities, formed mainly from the burning of coal and fossil fuels. It is a mixture of fog, smoke particles, ash and sulphur dioxide; the SO2 oxidises and combines with water droplets to form sulphuric acid aerosol: 2SO2 + O2 --> 2SO3, SO3 + H2O --> H2SO4. It forms especially in cold, damp, still weather under temperature inversion, and it causes respiratory diseases and damages buildings.

Question: 14. 4. What are the main gases responsible for green house effect?

Answer: The main greenhouse gases are carbon dioxide CO2 (from burning fossil fuels and deforestation), methane CH4 (from decaying matter, livestock and rice fields), nitrous oxide N2O (from fertilisers), water vapours, ozone O3 and chlorofluorocarbons CFCs. These gases are transparent to incoming sunlight but absorb the infrared heat radiation emitted by the Earth, keeping the atmosphere warm; their increase causes global warming.

Question: 15. 5. Explain four fundamental methods for the testing of waste water.

Answer: Four fundamental tests of waste water are: (1) pH measurement, which shows whether the water is acidic or alkaline compared with safe limits; (2) BOD (biochemical oxygen demand), the oxygen used by micro-organisms to decompose organic matter in a fixed time - higher BOD means more organic pollution; (3) COD (chemical oxygen demand), the oxygen needed to chemically oxidise the pollutants with a strong oxidising agent; and (4) determination of dissolved solids and specific pollutants such as heavy metals, oil and grease, so the water can be compared with WHO and environmental standards before discharge.

Descriptive Questions

Question: 16. 1. What is the effect of acid rain on human health and what measures can be taken to prevent acid rain?

Answer: Acid rain harms human health mainly indirectly: it dissolves toxic metals like aluminium, lead and mercury from soil into drinking water, irritates the eyes, skin and respiratory system through its precursor gases

SO2 and NOx, and damages crops and water supplies on which people depend. It also corrodes buildings, monuments and metal structures. Prevention measures include reducing SO2 and NOx emissions by using low-sulphur fuels, fitting scrubbers in factory chimneys and catalytic converters in vehicles, shifting to cleaner and renewable energy sources, and enforcing environmental laws through agencies like the EPA.

Question: 17. 2. What is the main cause of Global warming? How does it effect on weather pattern?

Answer: The main cause of global warming is the increased emission of greenhouse gases, chiefly carbon dioxide from the burning of coal, oil and gas in industries, vehicles and power stations, increased by deforestation which removes the trees that absorb CO2. The trapped heat raises the average temperature of the Earth. This disturbs weather patterns: heatwaves become more frequent, rainfall becomes irregular with more floods and droughts, glaciers and polar ice melt raising sea level, and storms and extreme weather events become stronger and less predictable.

Question: 18. 3. Describe the chemistry involves due to the presence of oxides of nitrogen and sulphur in the troposphere.

Answer: In the troposphere nitrogen and its oxides react as: N2 + O2 --lightning/engine heat--> 2NO, and 2NO + O2 --> 2NO2; NO2 with water and sunlight forms nitric acid and ozone, contributing to acid rain and photochemical smog, e.g. 4NO2 + 2H2O + O2 --> 4HNO3. Sulphur in coal and oil burns to SO2: S + O2 --> SO2, which oxidises further: 2SO2 + O2 --> 2SO3, and SO3 + H2O --> H2SO4 (sulphuric acid), or SO2 + H2O --> H2SO3. These acids return as acid rain, while NOx with hydrocarbons and

ozone together produce smog that harms lungs, crops and materials.

Question: 19. 4. What is Green house effect? How does human activities contributes to the enhancement of the green house effect?

Answer: The greenhouse effect is the warming of the Earth atmosphere by gases that allow sunlight to reach the surface but absorb the infrared (heat) radiation re-emitted by the Earth, like the glass of a greenhouse; in natural amounts it keeps the Earth warm enough for life. Human activities enhance it: burning fossil fuels in industries, vehicles and power plants releases huge amounts of CO2, deforestation reduces CO2 absorption, agriculture and livestock release methane and nitrous oxide, and industries release CFCs. The enhanced effect traps more heat and causes global warming and climate change.

Question: 20. 5. What are ozone depletion substances (ODS)? What human activities have contributed to ozone depletion in the stratosphere?

Answer: Ozone-depleting substances (ODS) are man-made chemicals that release chlorine or bromine in the stratosphere and destroy ozone, mainly chlorofluorocarbons (CFCs), halons, carbon tetrachloride, methyl chloroform and HCFCs. Human activities that contributed include the wide use of CFCs as refrigerants in refrigerators and air conditioners, as propellants in aerosol sprays and in foam blowing, halons in fire extinguishers, and nitrogen oxides from jet aircraft and fertilisers. Because of the resulting ozone hole and increased UV radiation, the Montreal Protocol has phased out most ODS and replaced them with safer alternatives like HFCs and hydrocarbons.

Question: 21. 6. What is a catalytic converter? What are the main pollution

targeted by catalytic converter?

Answer: A catalytic converter is a device fitted in the exhaust system of automobiles, containing a honeycomb coated with platinum, palladium and rhodium catalysts, which converts harmful exhaust gases into harmless ones at high temperature. The main pollutants it targets are carbon monoxide CO (converted to CO2), unburnt hydrocarbons (converted to CO2 and H2O) and nitrogen oxides NOx (converted to N2 and O2): for example 2CO + O2 --> 2CO2 and 2NO --> N2 + O2. It requires unleaded petrol, because lead poisons the catalyst.

Chapter 13: Spectroscopy

Multiple Choice Questions

1. (i) The energy of IR radiations is:

✓ (a) Higher than radio waves

(b) Higher than U.V waves

(c) Lower than micro waves

(d) Higher than x-rays

Answer: (a) Higher than radio waves - In the electromagnetic spectrum, energy increases in the order radio waves, microwaves, infrared, visible, UV, X-rays, gamma rays (E = hv, so higher frequency means higher energy). IR radiation therefore has energy higher than radio waves and microwaves, but lower than UV waves and X-rays.

2. (ii) The fingerprint region of an IR spectrum typically found in the range of:

(a) 1500-4000cm-1

(b) 2000-4000cm-1

✓ (c) 1500-500cm-1

(d) 4000-8000cm-1

Answer: (c) 1500-500cm-1 - The region of the IR spectrum from about 1500 to 500 cm-1 contains many bending and skeletal vibrations and gives a complex pattern that is unique for each molecule, like a fingerprint of a person. It is used to identify a compound by comparing its spectrum with that of a known sample.

Question: 3. (iii) Which unit of measurement is used for the wave number

of IR spectrum: (a) Nanometer (nm) (b) Angstrom (A) (c) Centimeter-1 (cm-1) (d) Hertz (Hz)

Answer: (c) Centimeter-1 (cm-1) - In IR spectroscopy frequency is expressed as wavenumber, the number of waves per centimetre, whose unit is reciprocal centimetre, cm-1. The useful IR range is about 4000-670 cm-1, while nanometre and angstrom are units of wavelength used in UV-visible spectroscopy.

4. (iv) Electronic excitation occurs in electromagnetic spectrum if the molecule absorbs:

(a) I.R radiations

✓ (b) U.V-visible radiations

(c) Radio waves

(d) Micro waves

Answer: (b) U.V-visible radiations - UV and visible radiations have enough energy to raise electrons of a molecule from a lower energy level to a higher one (electronic excitation or electronic transition). IR radiation only increases bond vibrations and radio waves only change nuclear spin, so neither causes electronic excitation.

5. (v) In NMR spectroscopy, the hydroxyl proton of C2H5OH appears as a broad singlet around:

(a) 1 - 2 ppm

(b) 2 - 3 ppm

✓ (c) 4 - 5 ppm

(d) 6 - 7 ppm

Answer: (c) 4 - 5 ppm - The -OH proton of ethanol is deshielded by the electronegative oxygen and by hydrogen bonding, so it appears downfield

as a broad singlet at about 4-5 ppm in the proton NMR spectrum as given in the book. Its exact position varies with concentration and solvent because of exchange of the OH proton.

6. (vi) Atomic absorption spectrum is represented by:

✓ (a) Dark lines against bright background

(b) Bright lines against dark background

(c) Bright lines against bright background

(d) Dark lines against dark background

Answer: (a) Dark lines against bright background - In an absorption spectrum, atoms absorb light of their characteristic wavelengths from a continuous source, so those wavelengths are missing and appear as dark lines on a bright continuous background. The opposite - bright lines on a dark background - is an emission spectrum.

7. (vii) Infrared spectroscopy is a technique use to determine _____ in the given organic molecule:

(a) Double and triple bonds

(b) Mass to charge ratio

✓ (c) Functional group

(d) Conjugated system

Answer: (c) Functional group - Each functional group absorbs IR radiation at a characteristic wavenumber (for example C=O at about 1680-1740 cm-1 and O-H at about 3600-3650 cm-1). By noting which absorptions are present, IR spectroscopy identifies the bonds and functional groups present in an organic molecule.

8. (viii) Highest U.V-visible absorption energy require for the transition of:

(a) sigma to sigma*

(b) pi to pi*

(c) n to sigma*

(d) n to pi*

Answer: (a) sigma to sigma* - The sigma to sigma* transition requires the most energy because sigma bonds are very strong and their electrons are held tightly, so it absorbs at the shortest wavelength in the far UV. The energy order is sigma to sigma* highest, then n to sigma*, then pi to pi*, and n to pi* lowest.

9. (ix) NMR spectroscopy is carried out if radiations interact with molecules in high magnetic field:

(a) U.V-visible rays

✓ (b) Radio rays

(c) X-rays

(d) Infrared rays

Answer: (b) Radio rays - NMR spectroscopy uses low-energy radiofrequency (radio) radiations. In a strong magnetic field, nuclei such as 1H absorb radio waves at a characteristic resonance frequency, and this absorption is recorded to give information about the environment of the nuclei.

10. (x) In which of the following spectroscopy technique D2O can be used as solvent:

(a) Mass spectroscopy

✓ (b) NMR spectroscopy

(c) Atomic absorption spectroscopy

(d) I.R spectroscopy

✓

Answer: (b) NMR spectroscopy - D2O (heavy water) is used as a solvent in NMR spectroscopy because deuterium does not give a proton NMR signal, so it does not hide the signals of the sample. It is also used to identify -OH and -NH protons, which exchange with deuterium and their peaks disappear from the spectrum.

Short Questions

Question: 11. 1. What types of nuclei are detected in proton NMR spectroscopy?

Answer: Proton NMR detects nuclei that have spin, principally the hydrogen nucleus 1H (a single proton) present in the sample - hence it is also called 1H NMR. Only nuclei with an odd number of protons or neutrons, such as 1H and 13C, have a magnetic spin and can absorb radio waves in a magnetic field and be detected. In proton NMR the signals of hydrogen atoms in different chemical environments are recorded separately.

Question: 12. 2. Name the components which represents x-axis and y-axis of a proton NMR spectrum.

Answer: In a proton NMR spectrum the x-axis represents the chemical shift, delta, measured in ppm (parts per million), increasing from right (0 ppm, the TMS reference) to left (downfield, up to about 10-12 ppm). The y-axis represents the intensity or strength of absorption (signal height/area), which is proportional to the number of equivalent protons producing that signal. Thus peak position tells the environment of the protons and peak area tells how many protons are present.

Question: 13. 3. Differentiate between atomic absorption and emission

spectroscopy.

Answer: In atomic emission spectroscopy the sample is heated or excited so that its atoms emit light of characteristic wavelengths on returning to lower energy levels, giving bright lines on a dark background. In atomic absorption spectroscopy light from a lamp is passed through the vapour of the sample and the atoms absorb their characteristic wavelengths, giving dark lines on a bright background. Emission measures the light given out and is used mostly for qualitative detection, while absorption measures the light absorbed and is widely used for quantitative determination of metals.

Question: 14. 4. What is the purpose of U.V-visible spectroscopy? What is its applications in chemistry and biology?

Answer: The purpose of UV-visible spectroscopy is to study molecules that absorb UV or visible light through electronic transitions, and thereby to identify conjugated systems and measure the concentration of a substance from its absorbance. In chemistry it is used for structure determination, studying reaction kinetics, detecting impurities and in quantitative analysis and quality control of drugs. In biology it is used to measure proteins and nucleic acids (DNA absorbs at 260 nm and proteins at 280 nm), to study enzymes and to estimate the concentration of biological samples.

Descriptive Questions

Question: 15. 1. What information about the structure of a molecule we can get from mass spectroscopy? Give the applications of mass spectroscopy.

Answer: Mass spectroscopy gives the accurate relative molecular mass of

a compound from its molecular ion peak (M+), from which its molecular formula can be worked out. The fragmentation pattern - the smaller peaks at different mass-to-charge (m/z) values - shows how the molecule breaks up, which reveals the groups and arrangement of atoms present and can distinguish between isomers. Its applications include identification of unknown compounds, structure determination, purity checking, detection of drugs and forensic materials, isotope abundance measurement, and quality control in the pharmaceutical and food industries.

Question: 16. 2. What is proton NMR spectroscopy? How does it work? Give its applications.

Answer: Proton NMR (1H NMR) spectroscopy is the study of the absorption of radio waves by hydrogen nuclei of a molecule placed in a strong magnetic field. The protons act like tiny magnets and align with or against the field; when they absorb radiofrequency energy of the right frequency they flip to the higher energy state (resonance). Protons in different environments absorb at slightly different frequencies (chemical shift, in ppm, with TMS as reference at 0 ppm), and the number of signals, their splitting and their areas reveal the structure of the molecule. It is applied in determining the structure of organic compounds, identifying ethanol and other substances, studying proteins, medical imaging (MRI) and quality control.

Question: 17. 3. Explain lambda max with the help of U.V-visible spectrum of ethanol (C2H5OH).

Answer: Lambda max is the wavelength at which a substance shows the maximum absorption of UV-visible radiation in its spectrum; it is a characteristic property of the compound and the transition involved. In the

UV-visible spectrum of ethanol shown in the book (absorbance plotted against wavelength in nm, 200-500 nm), the curve rises to a single strong peak with maximum absorbance of about 1.5 at about 270-280 nm and then falls to nearly zero above 300 nm. The wavelength at the top of this peak is the lambda max of ethanol, arising from the n to sigma* transition of the lone-pair electrons of the -OH oxygen, and ethanol shows no significant absorption in the visible region.

Question: 18. 4. Explain the graph of proton NMR of ethanol (C2H5OH) proton peaks of OH, CH2 and CH3.

Answer: The proton NMR graph of ethanol shows three signals because ethanol has three kinds of protons: CH3, CH2 and OH, in the ratio 3:2:1. Peak a at about 1.2 ppm is the CH3 group, split into a triplet by the two neighbouring CH2 protons, and it has the largest area (3 protons). Peak c at about 3.7 ppm is the CH2 group, split into a quartet by the three CH3 protons, with area for 2 protons; it is further downfield because it is directly attached to oxygen. Peak b at about 4.6 ppm is the single OH proton, appearing as a broad singlet with the smallest area (1 proton), most deshielded because it is bonded to electronegative oxygen. Together the positions, splitting and areas of a, b and c confirm the structure CH3-CH2-OH.